Data Analysis Portfolio

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PROFESSIONAL BACKGROUND:

After completing a comprehensive Data Analysis course, I gained hands-on experience with essential tools and techniques used to extract, analyze, and visualize complex data. I developed proficiency in tools such as Python, tableau, SQL, and Excel, applying them to real-world datasets to derive actionable insights. Through the course, I mastered data cleaning, statistical analysis, and the creation of interactive visualizations that helped tell compelling data-driven stories.

I also worked on several projects, allowing me to refine my problem-solving skills and ability to communicate findings to non-technical stakeholders. The course not only equipped me with the technical expertise to manipulate large datasets but also honed my ability to identify patterns and trends that inform strategic business decisions. I am now eager to apply my skills in a professional setting, where I can continue to grow and contribute to data-driven decision-making processes.

Table Of Contents

Professional Background ---------------- 2

Table of Contents ---------------------- 3

Instagram User Analytics

Project description---------------------6

The Problem --------------------------- 6

Design ------------------------------- 7

Findings ----------------------------- 7-11

Conclusion----------------------------11

Hiring Process Analytics

Project description---------------------12

The Problem --------------------------- 12

Design ------------------------------- 12

Findings -----------------------------13-16

Conclusion --------------------------- 16

IMDB Movie Analysis

Project description------------------------------17
The Problem -- 17
Design --- 18
Findings ---18-23
Conclusion -- 23

Bank Loan Case Study

Project description------------------------------24
The Problem -- 24
Design --- 25
Findings ---25-30
Conclusion -- 30

Impact of Car Features

Project description ---------------------- 31

The Problem ---------------------- 31

Design ---------------------- 32

Findings ---------------------- 33-39

Conclusion ---------------------- 40

ABC Call Volume Trend

Project description ---------------------- 41

The Problem ---------------------- 41

Design ---------------------- 42

Findings ---------------------- 42-45

Conclusion ---------------------- 45

Appendix ---------------------- 46

Instagram User Analytics

Project Description:

User analysis involves tracking how users engage with a digital product, such as a software application or a mobile app. The insights derived from this analysis can be used by various teams within the business. For example, the marketing team might use these insights to launch a new campaign, the product team might use them to decide on new features to build, and the development team might use them to improve the overall user experience.

Problem:

you are working with the product team of Instagram and the product manager asked You to provide insights on questions asked by management team

Design:

Steps taken to clean the data

- First importing the datasets provided and merging them.
- Then removing the duplicates and the blank cells.
- Improving the headers of each column with proper values.
- Finding and replacing the data with correct values.

Tools used for visualization: SQL

Findings:

1. **Loyal User Reward:** The marketing team wants to reward the most loyal users, i.e., those who have been using the platform for the longest time.

Identify the five oldest users on Instagram from the provided database.

Result Grid | Filter Rows:

id	username	created_at
180	Darby_Herzog	2016-05-06 00:14:21
80	Darby_Herzog	2016-05-06 00:14:21
167	Emilio_Bernier52	2016-05-06 13:04:30
67	Emilio_Bernier52	2016-05-06 13:04:30
63	Elenor88	2016-05-08 01:30:41
NULL	NULL	NULL

Inactive User Engagement: The team wants to encourage inactive users to start posting by sending them promotional emails.

Identify users who have never posted a single photo on Instagram.

username
Aniya_Hackett
Kasandra_Homenick
Jaclyn81
Rocio33
Maxwell.Halvorson
Tierra.Trantow
Pearl7
Ollie_Ledner37
Mckenna17
David.Osinski47
Morgan.Kassulke
Linnea59

Contest Winner Declaration: The team has organized a contest where the user with the most likes on a single photo wins.
Determine the winner of the contest and provide their details to the team.

username	id	image_url	total
Zack_Kemmer93	145	https://jarret.name	48

Hashtag Research: A partner brand wants to know the most popular hashtags to use in their posts to reach the most people.
Identify and suggest the top five most commonly used hashtags on the platform.

tag_name	total
smile	59
beach	42
party	39
fun	38
concert	24

Ad Campaign Launch: The team wants to know the best day of the week to launch ads.

Determine the day of the week when most users register on Instagram. Provide insights on when to schedule an ad campaign.

Result Grid

day	total
Thursday	32
Sunday	32
Friday	30

User Engagement: Investors want to know if users are still active and posting on Instagram or if they are making fewer posts.

Calculate the average number of posts per user on Instagram. Also, provide the total number of photos on Instagram divided by the total number of users.

Result Grid

avg
1.2850

Bots & Fake Accounts: Investors want to know if the platform is crowded with fake and dummy accounts.

Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.

Result Grid | Filter Rows:

username	num_likes
Aniya_Hackett	257
Jaclyn81	257
Rocio33	257
Maxwell.Halvorson	257
Ollie_Ledner37	257
Mckenna17	257
Duane60	257
Julien_Schmidt	257
Mike.Auer39	257
Nia_Haag	257
Leslie67	257
Janelle.Nikolaus81	257

Conclusion: The goal of this project is to use my SQL skills to extract meaningful insights from the data. My findings could potentially influence the future development of one of the world's most popular social media platforms.

Hiring Process Analytics:
Project Description

The hiring process is a crucial function of any company, and understanding trends such as the number of rejections, interviews, job types, and vacancies can provide valuable insights for the hiring department.

Problem:

With given dataset containing records of previous hires. My Task is to analyze this data and answer certain questions that can help the company improve its hiring process.

Design:

Steps taken to clean the data

- First importing the datasets provided and merging them.
- Then removing the duplicates and the blank cells.

- Improving the headers of each column with proper values.
- Finding and replacing the data with correct values.

Tools used for visualization: Microsoft Excel

Findings:

Hiring Analysis: The hiring process involves bringing new individuals into the organization for various roles.

Determine the gender distribution of hires. How many males and females have been hired by the company?

GENDERS	HIRED
MALES	2563
FEMALES	1856

Salary Analysis: The average salary is calculated b adding up the salaries of a group of employees and then dividing the total by the number of employees.

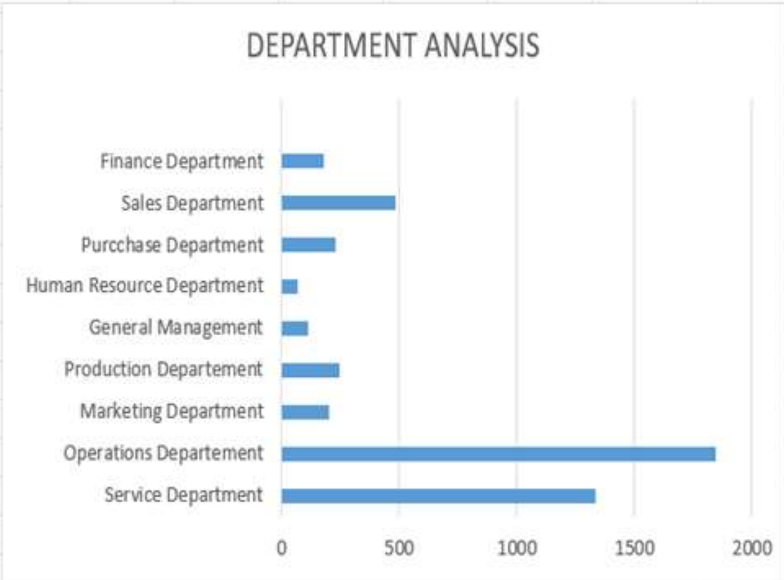
Average salary offered	49983.03

Salary Distribution: Class intervals represent ranges of values, in this case, salary ranges. The class interval is the difference between the upper and lower limits of a class.

class interval	Frequency
0-10000	678
10001-20000	732
20001-30000	711
30001-40000	710
40001-50000	782
50001-60000	750
60001-70000	698
70001-80000	734
80001-90000	711
90001-100000	659
100001 -200000	1
200001-300000	1
300001-400000	1
GRAND TOTAL	**7168**

Departmental Analysis: Visualizing data through charts and plots is a crucial part of data analysis. Use a pie chart, bar graph, or any other suitable visualization to show the proportion of people working in different departments.

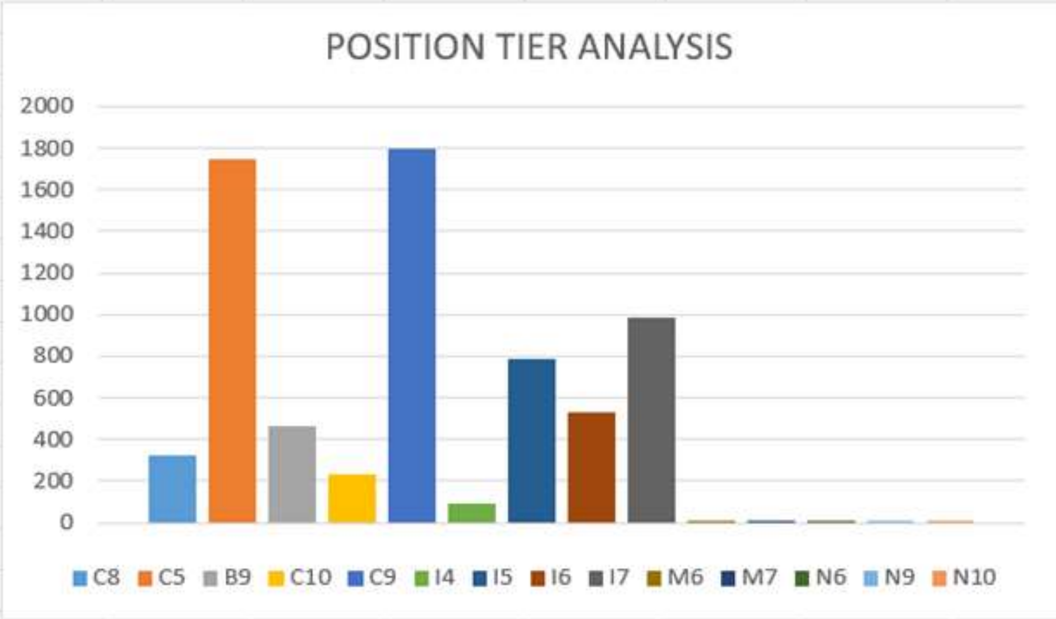
Department	Hired count
Service Department	1332
Operations Departement	1843
Marketing Department	202
Production Departement	246
General Management	113
Human Resource Department	70
Purcchase Department	230
Sales Department	485
Finance Department	176



Position Tier Analysis: Different positions within a company often have different tiers or levels.

Use a chart or graph to represent the different position tiers within the company. This will help you understand the distribution of positions across different tiers.

Postion Tire	No of People
C8	320
C5	1747
B9	463
C10	232
C9	1792
I4	88
I5	787
I6	527
I7	982
M6	3
M7	1
N6	1
N9	1
N10	1



Conclusion:

The goal of this project is to use my knowledge of statistics and Excel to draw meaningful conclusions about the company's hiring process. my insights could potentially help the company improve its hiring process and make better hiring decisions in the future.

IMDB Movie Analysis

Project Description:

Analysing IMDB could potentially provide insights, patterns and recommendations so that it could influence future projects and could possibly thrive for growth related to projects

Problem:

The dataset provided is related to IMDB Movies. A potential problem to investigate could be: "What factors influence the success of a movie on IMDB?"
Here, success can be defined by high IMDB ratings. The impact of this problem is significant for movie producers, directors, and investors who want to understand what makes a movie successful to make informed decisions in their future projects.

Design:

Steps taken to clean the data

- First importing the datasets provided and merging them.
- Then removing the duplicates and the blank cells.
- Improving the headers of each column with proper values.
- Finding and replacing the data with correct values.

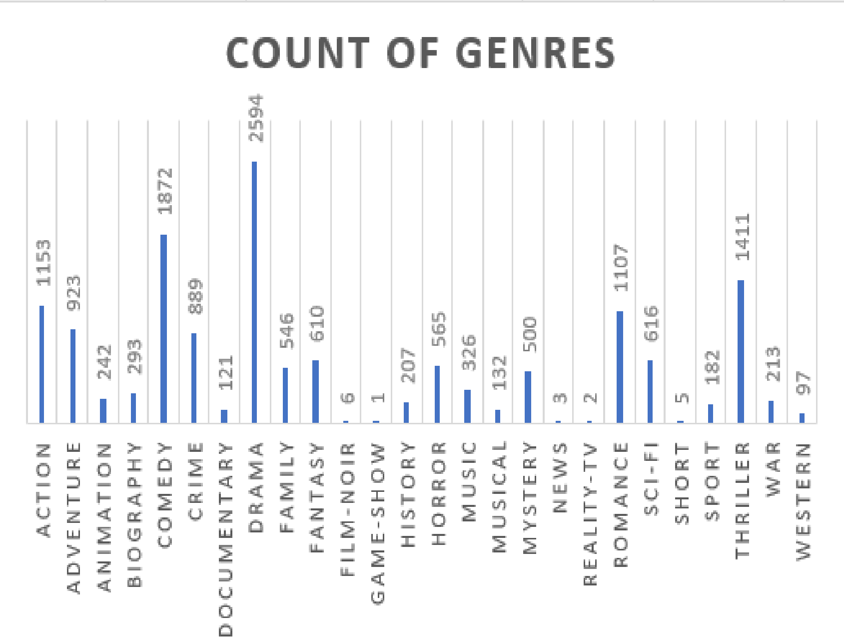
Tools used for visualization: Microsoft Excel

Findings:

Movie Genre Analysis: Analyse the distribution of movie genres and their impact on the IMDB score.

Determine the most common genres of movies in the dataset. Then, for each genre, calculate descriptive statistics

GENRES	COUNT OF GENRES	MEAN	MEDIAN	MODE	MAX	MIN	VARIANCE	STANDARD DEVIATION
Action	1153	6.239896	6.3	6.1	9.1	1.7	1.25179235	1.118835265
Adventure	923	6.44117	6.6	6.7	8.9	1.9	1.279604703	1.131196138
Animation	242	6.576033	6.7	6.7	8.6	1.7	1.298676314	1.139594803
Biography	293	7.150171	7.2	7	8.9	4.5	0.52202908	0.722515799
Comedy	1872	6.195246	6.3	6.7	9.5	1.7	1.189656701	1.090713849
Crime	889	6.564792	6.6	6.6	9.3	2.4	1.053612597	1.02645633
Documenta	121	7.180165	7.4	7.5	8.7	1.6	1.116269972	1.056536782
Drama	2594	6.763763	6.9	7.2	9.3	2	0.916526679	0.957353999
Family	546	6.245055	6.4	6.7	8.7	1.7	1.443837887	1.201598055
Fantasy	610	6.307049	6.4	6.7	8.9	1.7	1.347191607	1.160685835
Film-Noir	6	7.633333	7.65	#N/A	8.2	7.1	0.186666667	0.43204938
Game-Shov	1	2.9	2.9	#N/A	2.9	2.9	#DIV/0!	#DIV/0!
History	207	7.083575	7.2	7.5	8.9	2	0.788369682	0.887901843
Horror	565	5.84354	5.9	6.2	8.7	2.2	1.277959079	1.130468522
Music	326	6.457975	6.7	7.1	8.5	1.6	1.436043889	1.198350487
Musical	132	6.507576	6.7	7	8.5	2.1	1.502384918	1.225718123
Mystery	500	6.4864	6.6	6.6	8.6	2.2	1.189754549	1.090758703
News	3	7.533333	7.4	#N/A	8.1	7.1	0.263333333	0.513160144
Reality-TV	2	4.75	4.75	#N/A	6.6	2.9	6.845	2.61629509
Romance	1107	6.450587	6.5	6.5	8.6	2.1	0.992086002	0.996035141
Sci-Fi	616	6.281818	6.4	6.7	8.8	1.9	1.466075388	1.210816001
Short	5	6.38	6.5	#N/A	7.1	5.2	0.557	0.746324326
Sport	182	6.606044	6.8	7.2	8.7	2	1.214272661	1.101940407
Thriller	1411	6.314245	6.4	6.1	9	2.2	1.111619625	1.054333735
War	213	7.070423	7.1	7.1	8.6	2.7	0.765111613	0.874706587
Western	97	6.689691	6.8	6.5	8.9	3.8	1.086767612	1.042481468
Total	14616							

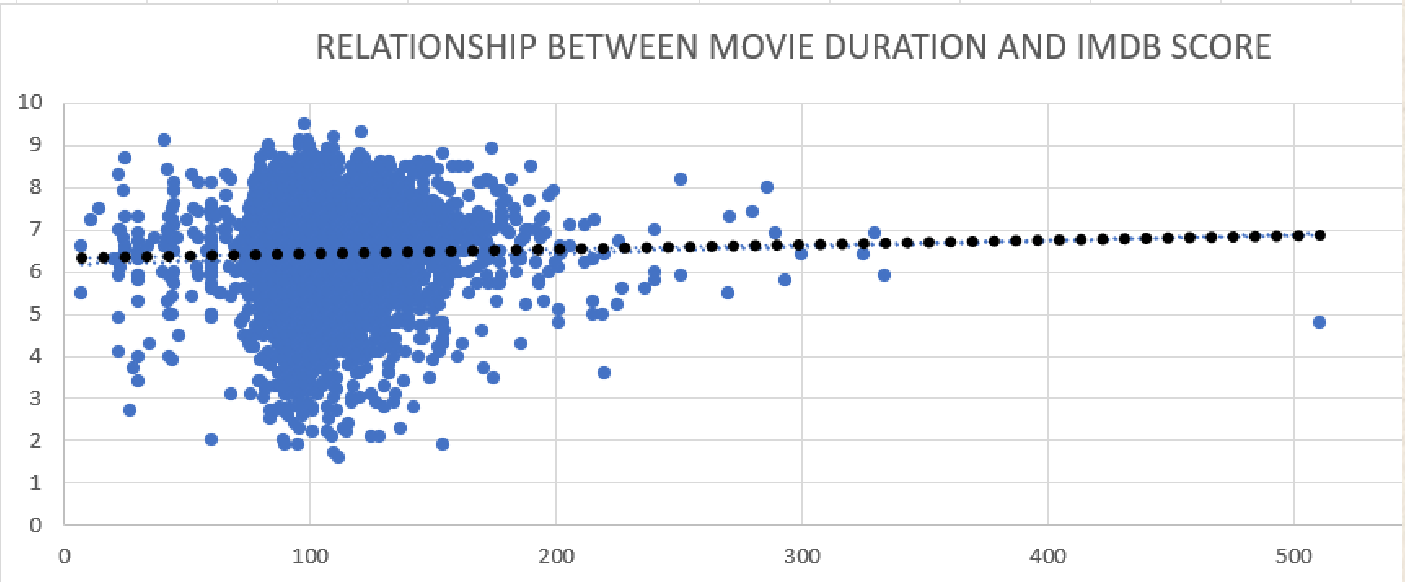


TOP GENRES	COUNTOF GENRES
Drama	2594
Comedy	1872
Thriller	1411
Action	1153
Romance	1107

Movie Duration Analysis: Analyze the distribution of movie durations and its impact on the IMDB score.

Analyze the distribution of movie durations and identify the relationship between movie duration and IMDB score.

MEAN	MEDIAN	STANDARD DEVIATION
107.2011	103	25.19744081



Language Analysis: Situation: Examine the distribution of movies based on their language. Determine the most common languages used in movies and analyze their impact on the IMDB score using descriptive statistics.

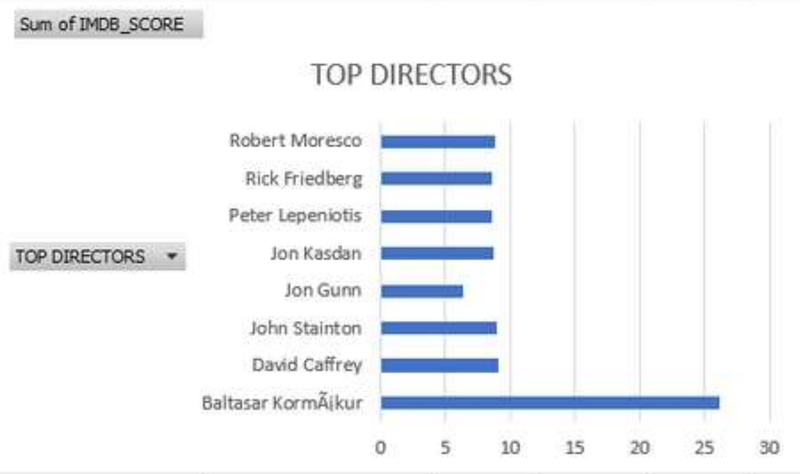
LANGUAGES	COUNT OF LANGUAGE		MEAN	MEDIAN	STANDARD DEVIATION
Aboriginal	2		7.5	7.5	0.141421356
Arabic	5		5.98	6.2	1.032956921
Aramaic	1		6.5	6.5	#DIV/0!
Bosnian	1		6.5	6.5	#DIV/0!
Cantonese	11		6.4	6.5	0.694262198
Chinese	3		7.46667	7.7	0.776745347
Czech	1		6.8	6.8	#DIV/0!
Danish	5		6.7	6.5	1.270826503
Dari	2		6.2	6.2	0.707106781
Dutch	4		6.55	6.45	1.410673598
Dzongkha	1		7.3	7.3	#DIV/0!
English	4704		6.43574	6.6	1.125716901
Filipino	1		6.5	6.5	#DIV/0!
French	73		6.53973	6.7	1.230259653
German	19		6.63158	6.7	0.797950591
Greek	1		7.2	7.2	#DIV/0!
Hebrew	5		6.46	6.1	0.918150314
Hindi	28		6.61429	6.85	1.347346991
Hungarian	1		6.4	6.4	#DIV/0!
Icelandic	2		5.65	5.65	1.626345597
Indonesian	2		6.75	6.75	0.070710678
Italian	11		6.28182	6.3	1.046726499
Japanese	18		6.28889	6.35	1.465641353
Kannada	1		5.5	5.5	#DIV/0!
Kazakh	1		7.3	7.3	#DIV/0!
Korean	8		6.6875	6.8	1.004898716
Mandarin	26		6.18462	6.15	1.179726174
Maya	1		6.3	6.3	#DIV/0!
Mongolian	1		7.9	7.9	#DIV/0!
None	2		7.2	7.2	0.565685425
Norwegian	4		7.575	7.35	0.567890835
Panjabi	1		5.3	5.3	#DIV/0!
Persian	4		6.775	6.95	0.543905629
Polish	4		6.9	6.8	0.346410162
Portuguese	8		6.8125	6.55	1.277763112
Romanian	2		7.5	7.5	0.707106781
Russian	11		6.13636	6.3	1.483423559
Slovenian	1		6.3	6.3	#DIV/0!
Spanish	40		6.645	6.5	1.085321647
Swahili	1		6.4	6.4	#DIV/0!
Swedish	5		6.4	6.4	1.034408043
Tamil	1		8	8	#DIV/0!
Telugu	1		5.7	5.7	#DIV/0!
Thai	3		6.3	6.4	0.173205081
Urdu	1		6.1	6.1	#DIV/0!
Vietnamese	1		6.3	6.3	#DIV/0!
Zulu	2		4.95	4.95	0.494974747
TOTAL	5031				

Director Analysis: Influence of directors on movie ratings.

Identify the top directors based on their average IMDB score and analyze their contribution to the success of movies using percentile calculations.

director_name	Average of imdb_score	Percentile
James Cameron	6.985714286	0.775
Gore Verbinski	6.95	0.764
Sam Mendes	6.571428571	0.554
Christopher Nolan	7.7625	0.955
Doug Walker	7.1	0.814
Andrew Stanton	7.333333333	0.895
Sam Raimi	6.436363636	0.472
Nathan Greno	7.8	0.956
Joss Whedon	6.925	0.757
David Yates	7.05	0.805
Zack Snyder	6.133333333	0.284
Bryan Singer	6.925	0.757
Marc Forster	5.8875	0.183
Gore Verbinski	6.95	0.765

TOP DIRECTORS	Sum of IMDB_SCORE
Baltasar KormÃ¡kur	26.2
David Caffrey	9.1
John Stainton	9
Jon Gunn	6.333333333
Jon Kasdan	8.7
Peter Lepeniotis	8.6
Rick Friedberg	8.6
Robert Moresco	8.9



Budget Analysis: Explore the relationship between movie budgets and their financial success. Analyze the correlation between movie budgets and gross earnings, and identify the movies with the highest profit margin.

s	budget	profit	correlation	TOP 5 MOVIES	
505847	237000000	523505847	0.09647	movie_title	MOVIE WITH HIGHEST PROFIT USING MAX FUNCTION
177271	150000000	502177271	0.095506	Avatar	Avatar
672302	200000000	458672302	0.095413	Jurassic World	
935665	11000000	449935665	0.094802	Titanic	
949459	10500000	424449459	0.095536	Star Wars: Episode IV - A New Hope	
279547	220000000	403279547	0.096206	E.T. the Extra-Terrestrial	
783777	45000000	377783777	0.095383		
544677	115000000	359544677	0.0958		
316061	185000000	348316061	0.095807		
999255	78000000	329999255	0.095288		
024263	58000000	305024263	0.095468		
645577	130000000	294645577	0.0957		
784000	63000000	293784000	0.095552		
049635	76000000	292049635	0.095749		
123553	58800000	291323553	0.095889		
838870	94000000	286838870	0.096104		
471036	150000000	286471036	0.096156		
019252	94000000	283019252	0.095881		
125409	32500000	276625409	0.095931		
691196	55000000	274691196	0.096218		
158751	18000000	272158751	0.096429		
761243	18000000	267761243	0.096752		

Conclusion

Goal is to not just to answer questions but to provide insights that can drive decision-making. My analysis aim to provide actionable insights that can help stakeholders make informed decisions.

Bank Loan Case Study

Project Description:

The primary goal of this case study is to identify patterns that indicate if a customer will have difficulty paying their installments. This information can be used to make decisions such as denying the loan, reducing the amount of loan, or lending at a higher interest rate to risky applicants. The company wants to understand the key factors behind loan default so it can make better decisions about loan approval.

Problem:

Imagine you're a data analyst at a finance company that specializes in lending various types of loans to urban customers. Your company faces a challenge: some customers who don't have a sufficient credit history take advantage of this and default on their loans.

Your task is to use Exploratory Data Analysis (EDA) to analyze patterns in the data and ensure that capable applicants are not rejected.

Design:

Steps taken to clean the data

- First importing the datasets provided and merging them.
- Then removing the duplicates and the blank cells.
- Improving the headers of each column with proper values.
- Finding and replacing the data with correct values.

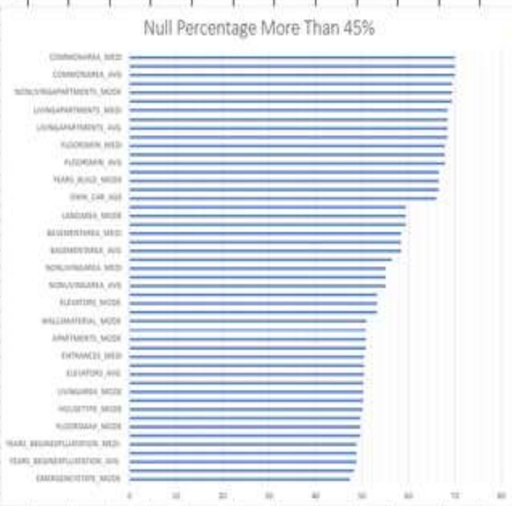
Tools used for visualization: Microsoft Excel

Findings:

Identify Missing Data and Deal with it Appropriately: As a data analyst, you come across missing data in the loan application dataset. It is essential to handle missing data effectively to ensure the accuracy of the analysis.

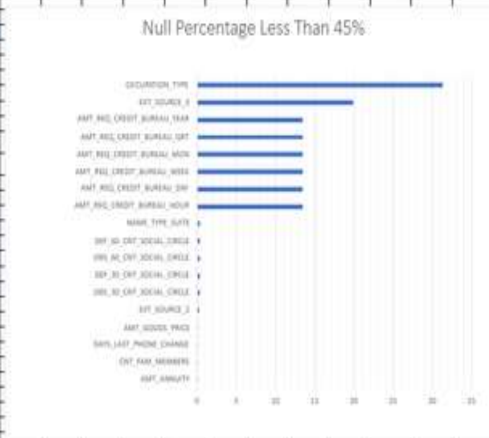
COLUMN HAVING NULL VALUE MORE THAN 45%

Column Name	Null Percentage
EMERGENCYSTATE_MODE	47.39894794
TOTALAREA_MODE	48.29898584
YEARS_BEGINEXPLUATATION_AVG	48.78897578
YEARS_BEGINEXPLUATATION_MODE	48.78897578
YEARS_BEGINEXPLUATATION_MEDI	48.78897578
FLOORSMAX_AVG	49.75099502
FLOORSMAX_MODE	49.75099502
FLOORSMAX_MEDI	49.75099502
HOUSETYPE_MODE	50.15100302
LIVINGAREA_AVG	50.2750055
LIVINGAREA_MODE	50.2750055
LIVINGAREA_MEDI	50.2750055
ELEVATORS_AVG	50.39100782
ENTRANCES_MODE	50.39100782
ENTRANCES_MEDI	50.39100782
APARTMENTS_AVG	50.77101542
APARTMENTS_MODE	50.77101542
APARTMENTS_MEDI	50.77101542
WALLSMATERIAL_MODE	50.91901838
ELEVATORS_AVG	53.30506606
ELEVATORS_MODE	53.30506606
ELEVATORS_MEDI	53.30506606
NONLIVINGAREA_AVG	55.1451029
NONLIVINGAREA_MODE	55.1451029
NONLIVINGAREA_MEDI	55.1451029
EXT_SOURCE_1	56.3431269
BASEMENTAREA_AVG	58.39916798
BASEMENTAREA_MODE	58.39916798
BASEMENTAREA_MEDI	58.39916798
LANDAREA_AVG	59.44318886
LANDAREA_MODE	59.44318886
LANDAREA_MEDI	59.44318886
OWN_CAR_AGE	[illegible]



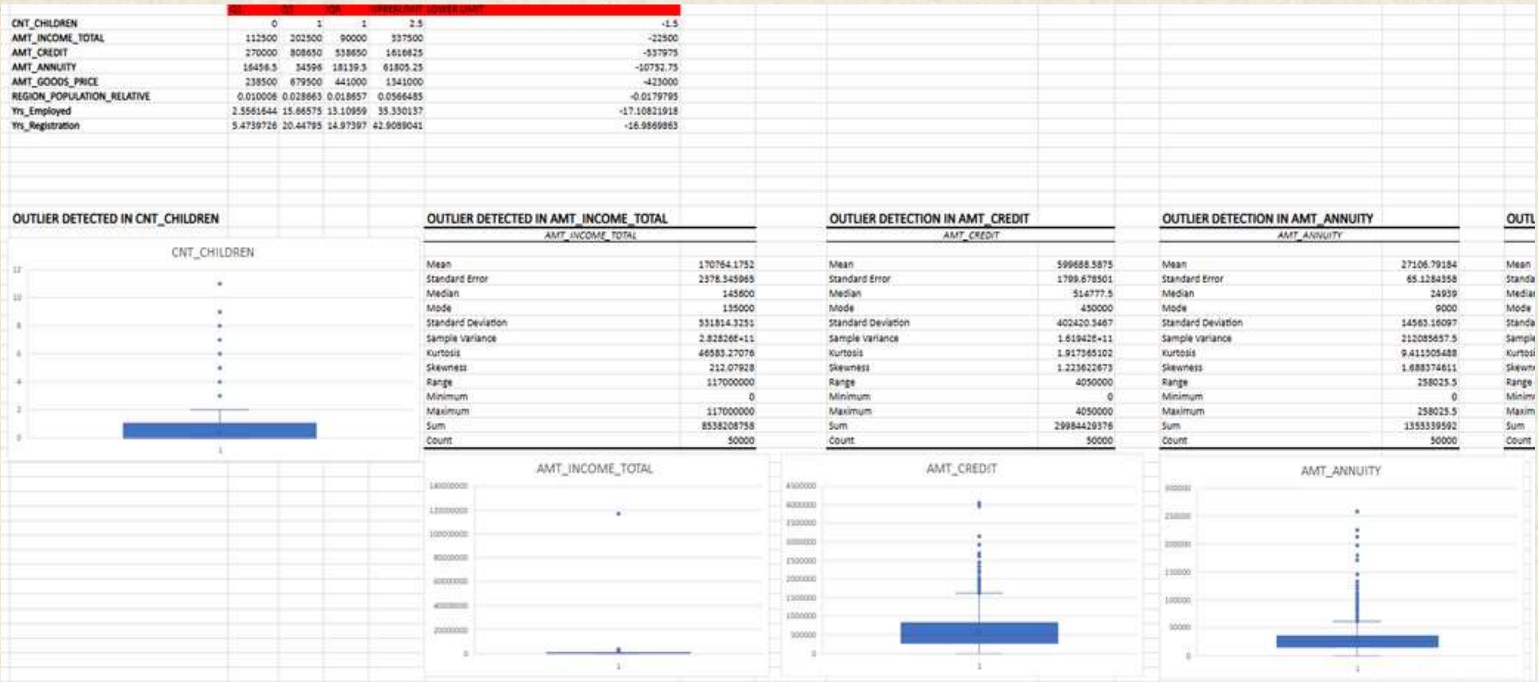
COLUMN HAING NULL VALUE LESS

Column Name	Null Percentage
AMT_ANNUITY	0.00200004
CNT_FAM_MEMBERS	0.00200004
DAYS_LAST_PHONE_CHANGE	0.00200004
AMT_GOODS_PRICE	0.07600152
EXT_SOURCE_2	0.25200504
OBS_30_CNT_SOCIAL_CIRCLE	0.33600672
DEF_30_CNT_SOCIAL_CIRCLE	0.33600672
OBS_60_CNT_SOCIAL_CIRCLE	0.33600672
DEF_60_CNT_SOCIAL_CIRCLE	0.33600672
NAME_TYPE_SUITE	0.58400768
AMT_REQ_CREDIT_BUREAU_HOUR	13.48826937
AMT_REQ_CREDIT_BUREAU_DAY	13.48826937
AMT_REQ_CREDIT_BUREAU_WEEK	13.48826937
AMT_REQ_CREDIT_BUREAU_MON	13.48826937
AMT_REQ_CREDIT_BUREAU_QRT	13.48826937
AMT_REQ_CREDIT_BUREAU_YEAR	13.48826937
EXT_SOURCE_3	19.88839777
OCCUPATION_TYPE	31.30882617



				BLANK VALUES	1	38	1	126	9944	168	168	
				MEAN	27107.37736	539060.0361	2.158946358	0.513823595	0.511881408	1.420782244	0.141819349	
				MEDIAN	24939	450000	2	0.565585366	0.53527625	0	0	
E_SUITE	OCCUPATION_TYPE	MODE FOR NAME_TYPE _SUITE			AMT_ANNUITY	AMT_GOODS_PRICE	CNT_FAM_MEMBERS	EXT_SOURCE_2	EXT_SOURCE_3	OBS_30_CNT_SOCIAL_CIRCLE	DEF_30_CNT_SOCIAL_CIRCLE	OBS_60_CNT
anied	Laborers	Unaccompanied	40435		24700.5	351000	1	0.262948593	0.13937578	2	2	
	Core staff	Family	6549		35698.5	1129500	2	0.622245775		1	0	
anied	Laborers	Spouse, partner	1849		6750	135000	1	0.555912083	0.729566691	0	0	
anied	Laborers	Children	542		29686.5	297000	2	0.65044169		2	0	
anied	Core staff	Other_A	137		21865.5	513000	1	0.322738287		0	0	
rtner	Laborers	0	0		27517.5	454500	2	0.354224732	0.621226338	0	0	
anied	Accountants	Other_B	259		41301	1395000	3	0.723999852	0.492060094	1	0	
anied	Managers	Group of people	36		42075	1530000	2	0.714279286	0.54065445	2	0	
					33826.5	913500	2	0.205747288	0.751723715	1	0	
anied	Laborers	Unaccompanied	is with highest score		20250	405000	1	0.746643629		2	0	
anied	Core staff				21177	652500	3	0.651862333	0.363945239	0	0	
					10678.5	135000	2	0.555183162	0.652896552	0	0	
anied	Laborers				5881.5	67500	2	0.715041819	0.176652579	0	0	
anied	Drivers				28966.5	697500	3	0.566906613	0.77008707	0	0	
anied	Laborers				32778	679500	2	0.642656205		0	0	
	Laborers				20160	247500	1	0.346633981	0.678567689	0	0	
anied	Drivers				26149.5	387000	2	0.23637784	0.062103038	0	0	
anied	Laborers				13500	270000	3	0.683513346		4	0	
	Laborers				7875	157500	1	0.706428403	0.556727426	8	0	

Identify Outliers in the Dataset: Outliers can significantly impact the analysis and distort the results. You need to identify outliers in the loan application dataset.



	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
CNT_CHILDREN	0	1	1	2.5	-1.5
AMT_INCOME_TOTAL	112500	202500	90000	337500	-22500
AMT_CREDIT	270000	808650	538650	1616625	-537975
AMT_ANNUITY	16456.5	34596	18139.5	61805.25	-10752.75
AMT_GOODS_PRICE	238500	679500	441000	1341000	-423000
REGION_POPULATION_RELATIVE	0.010006	0.028663	0.018657	0.0566485	-0.0179795
Yrs_Employed	2.5561644	15.66575	13.10959	35.330137	-17.10821918
Yrs_Registration	5.4739726	20.44795	14.97397	42.9089041	-16.9869863

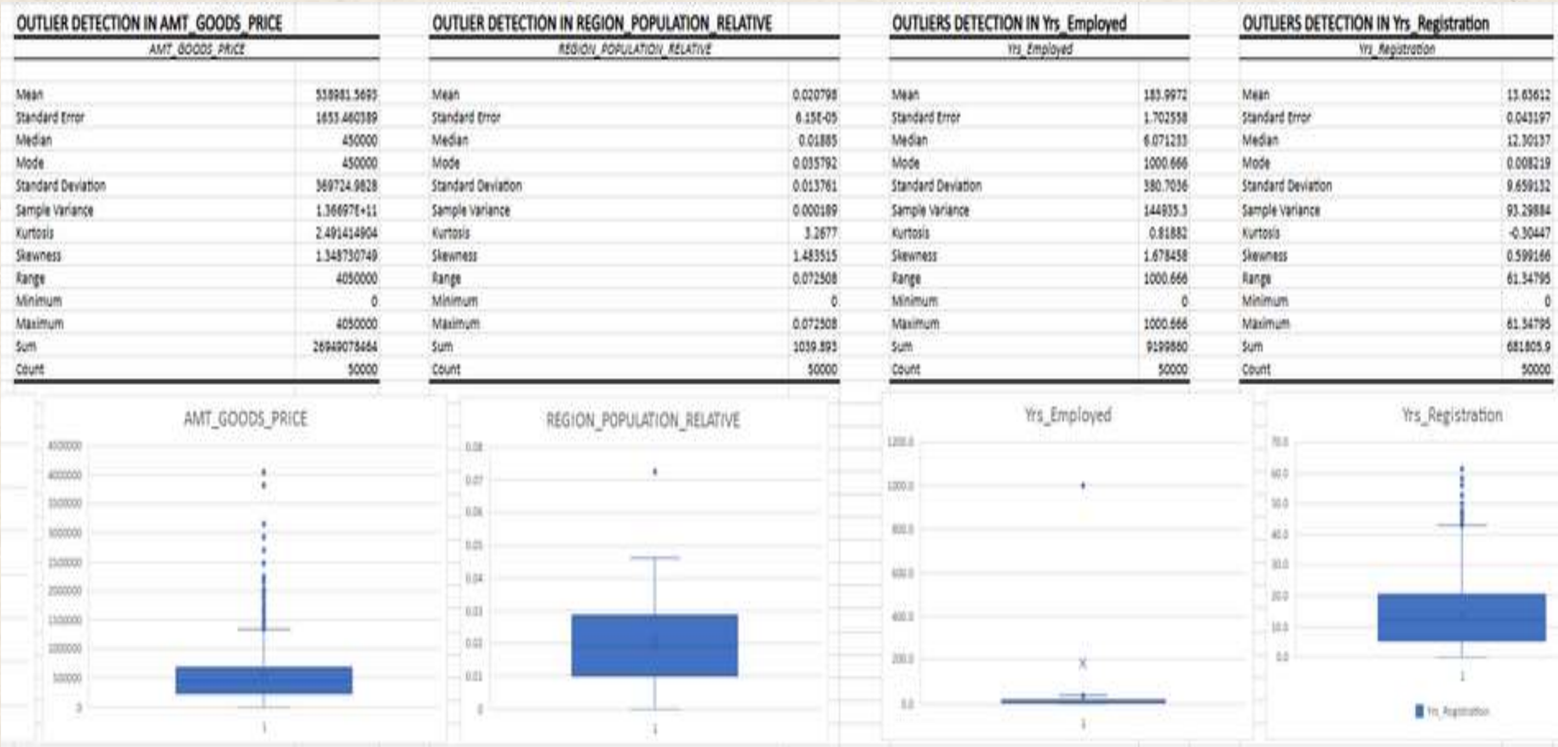
OUTLIER DETECTED IN CNT_CHILDREN

OUTLIER DETECTED IN AMT_INCOME_TOTAL

OUTLIER DETECTION IN AMT_CREDIT

OUTLIER DETECTION IN AMT_ANNUITY

	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY
Mean	170764.1752	599688.5875	27106.79184
Standard Error	2378.545965	1799.678501	65.1284358
Median	145800	514777.5	24939
Mode	135000	450000	9000
Standard Deviation	531814.3251	402420.5467	14563.16097
Sample Variance	2.82826E+11	1.61942E+11	212085657.5
Kurtosis	46583.27076	1.917365102	9.411505488
Skewness	212.07928	1.223622673	1.688374611
Range	117000000	4050000	258025.5
Minimum	0	0	0
Maximum	117000000	4050000	258025.5
Sum	8538208758	29984429376	1355339592
Count	50000	50000	50000



OUTLIER DETECTION IN AMT_GOODS_PRICE

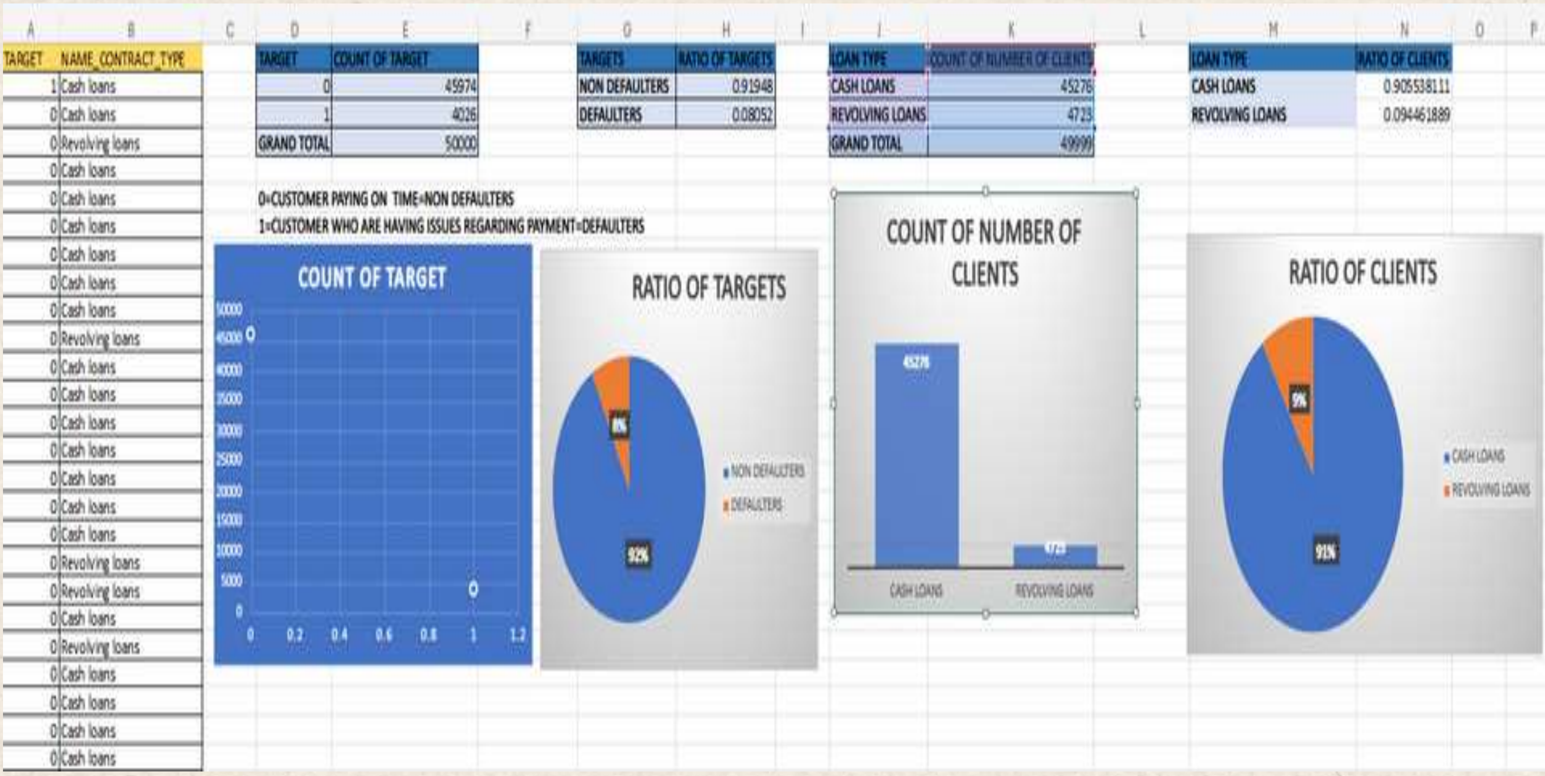
OUTLIER DETECTION IN REGION_POPULATION_RELATIVE

OUTLIERS DETECTION IN Yrs_Employed

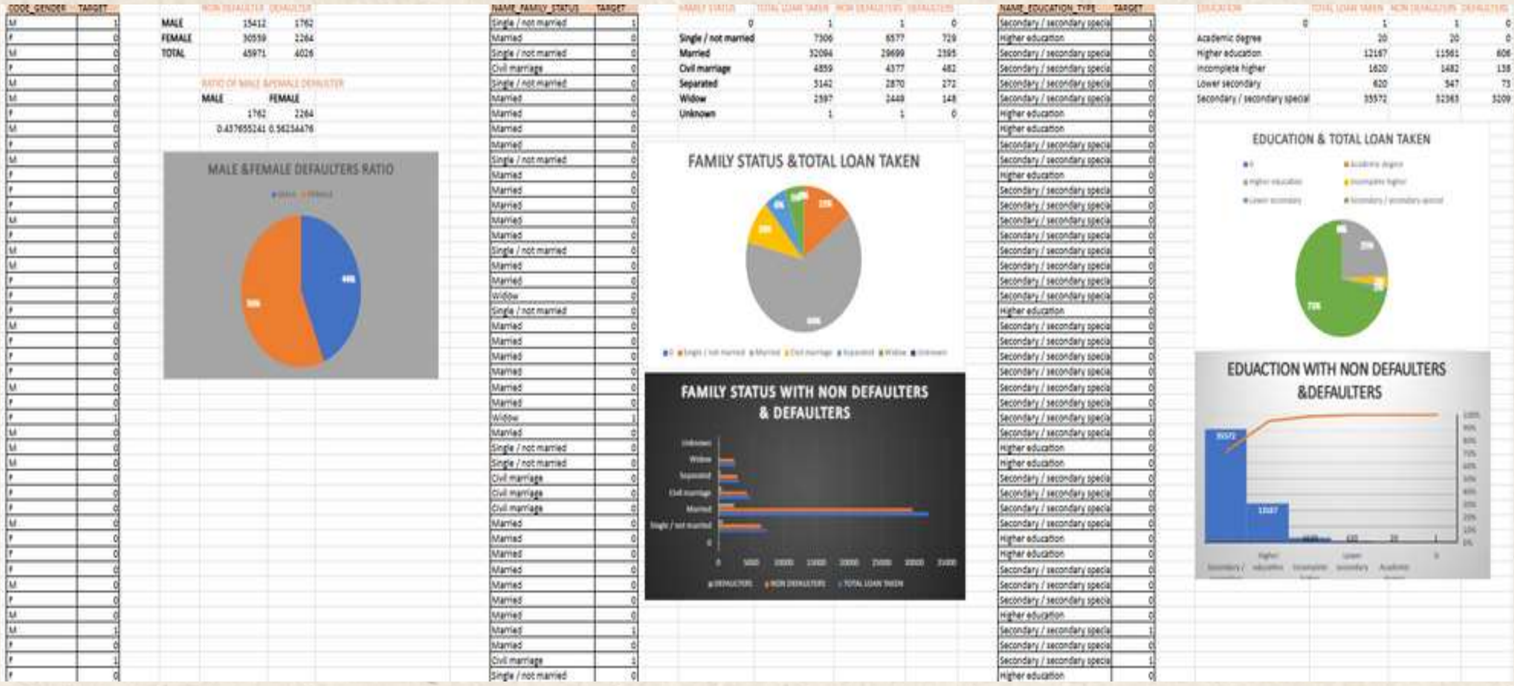
OUTLIERS DETECTION IN Yrs_Registration

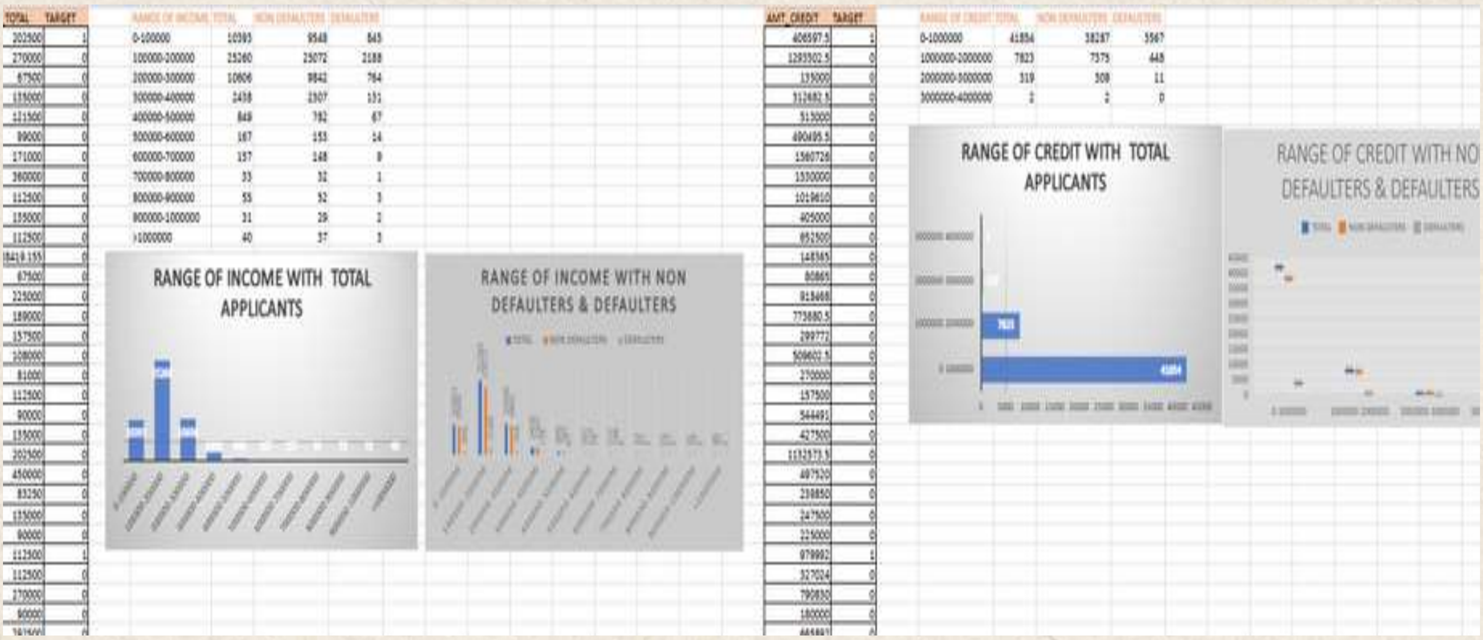
	AMT_GOODS_PRICE	REGION_POPULATION_RELATIVE	Yrs_Employed	Yrs_Registration
Mean	538981.5693	0.020798	183.9972	13.63612
Standard Error	1653.460389	6.15E-05	1.702558	0.043197
Median	450000	0.01885	6.071233	12.30137
Mode	450000	0.035792	1000.666	0.008219
Standard Deviation	369724.9828	0.013761	380.7036	9.659132
Sample Variance	1.36697E+11	0.000189	144935.3	93.29884
Kurtosis	2.491414904	3.2677	0.81882	-0.30447
Skewness	1.348730749	1.483515	1.678458	0.599166
Range	4050000	0.072508	1000.666	61.34795
Minimum	0	0	0	0
Maximum	4050000	0.072508	1000.666	61.34795
Sum	26949078464	1039.893	9199860	681805.9
Count	50000	50000	50000	50000

Analyze Data Imbalance: Data imbalance can affect the accuracy of the analysis, especially for binary classification problems. Understanding the data distribution is crucial for building reliable models.



Perform Univariate, Segmented Univariate, and Bivariate Analysis: To gain insights into the driving factors of loan default, it is important to conduct various analyses on consumer and loan attributes.





Identify Top Correlations for Different Scenarios

Understanding the correlation between variables and the target variable can provide insights into strong indicators of loan default.

	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	REGION_POPULATION_RELATIVE	Client_Age	Yrs_Employed	Yrs_id_publish	REGION_RATING_CLIENT
CNT_CHILDREN	1	0.00959224	0.004988715	-0.025537466	-0.32917562	-0.241532583	0.032137426	0.025956409
AMT_INCOME_TOTAL	0.00959224	1	0.069323857	0.029850463	-0.015977009	-0.03150712	-0.003493753	-0.038156309
AMT_CREDIT	0.004988715	0.069323857	1	0.095151984	0.05944281	-0.067723342	0.01228719	-0.100368442
REGION_POPULATION_RELATIVE	-0.025537466	0.029850463	0.095151984	1	0.032619652	-0.004143622	0.004404886	-0.532446194
Client_Age	-0.32917562	-0.015977009	0.05944281	0.032619652	1	0.621678475	0.270923789	-0.016477534
Yrs_Employed	-0.241532583	-0.03150712	-0.06772334	-0.004143622	0.621678475	1	0.272774474	0.034591974
Yrs_id_publish	0.032137426	-0.003493753	0.01228719	0.004404886	0.270923789	0.272774474	1	0.002466909
REGION_RATING_CLIENT	0.025956409	-0.038156309	-0.10036844	-0.532446194	-0.016477534	0.034591974	0.002466909	1

TOP CORRELATION

	Variable 1	Variable 2	Coorelation
1	Client_age	Yrs_Employec	0.621678475
2	Yrs_Employed	Yrs _id_publis	0.272774474
3	Yrs_id_publish	Client_age	0.270923789

	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	REGION_POPULATION_RELATIVE	Client_Age	Yrs_Employed	Yrs_id_publish	REGION_RATING_CLIENT
CNT_CHILDREN	1	0.010110177	0.007601905	-0.020359154	-0.2496732	-0.189773227	0.042360717	0.055515557
AMT_INCOME_TOTAL	0.010110177	1	0.015271444	-0.006180303	-0.009033662	-0.011758681	0.009122006	-0.012846697
AMT_CREDIT	0.007601905	0.015271444	1	0.067775624	0.142506035	0.018782223	0.043771901	-0.045024534
REGION_POPULATION_RELATIVE	-0.020359154	-0.006180303	0.067775624	1	0.016468731	0.007710059	0.005118563	-0.430032303
Client_Age	-0.2496732	-0.009033662	0.142506035	0.016468731	1	0.588242824	0.247896571	-0.045027112
Yrs_Employed	-0.189773227	-0.011758681	0.018782223	0.007710059	0.588242824	1	0.232661912	-0.009237108
Yrs_id_publish	0.042360717	0.009122006	0.043771901	0.005118563	0.247896571	0.232661912	1	-0.025335227
REGION_RATING_CLIENT	0.055515557	-0.012846697	-0.045024534	-0.430032303	-0.045027112	-0.009237108	-0.025335227	1

TOP COORELATION

	Variable 1	Variable 2	Coorelation
1	Client_age	Yrs_Employed	0.588242824
2	Yrs_id_publish	Client_age	0.247896571
3	Yrs_Employed	Yrs _id_publish	0.232661912

Conclusion:

Goal in this project is to use EDA to understand how customer attributes and loan attributes influence the likelihood of default.

Impact of Car Features

Project Description:

The automotive industry has been rapidly evolving over the past few decades, with a growing focus on fuel efficiency, environmental sustainability, and technological innovation. With increasing competition among manufacturers and a changing consumer landscape, it has become more important than ever to understand the factors that drive consumer demand for cars.

Problem:

For the given dataset, as a Data Analyst, the client has asked How can a car manufacturer optimize pricing and product development decisions to maximize profitability while meeting consumer demand?

Design:

Steps taken to clean the data

- First importing the datasets provided and merging them.
- Then removing the duplicates and the blank cells.
- Improving the headers of each column with proper values.
- Finding and replacing the data with correct values.

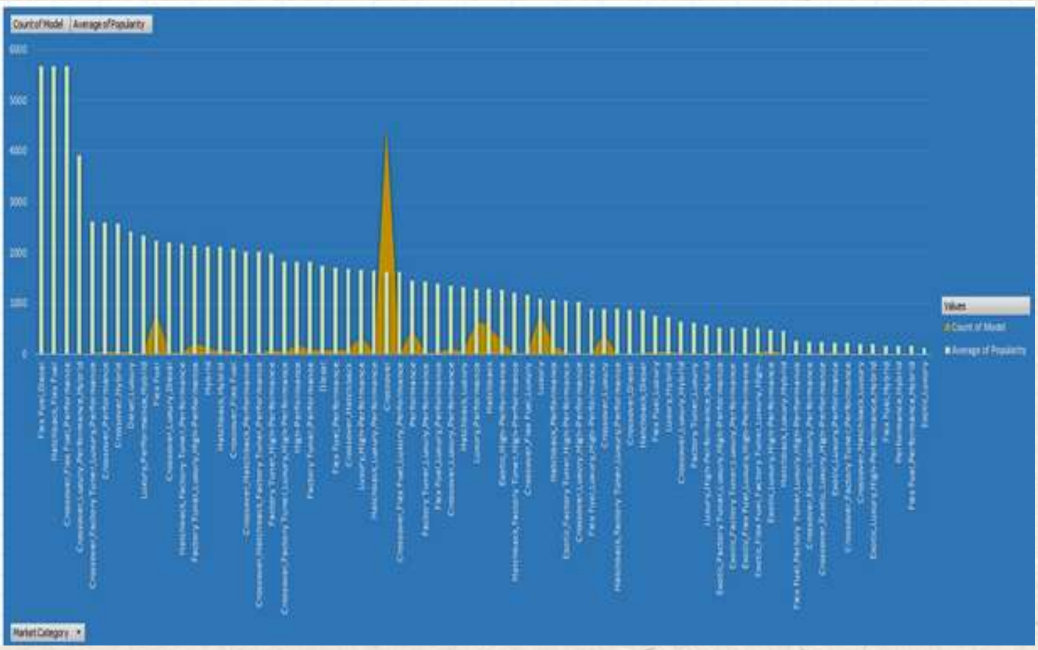
Tools used for visualization: Microsoft Excel

Findings:

How does the popularity of a car model vary across different market categories?

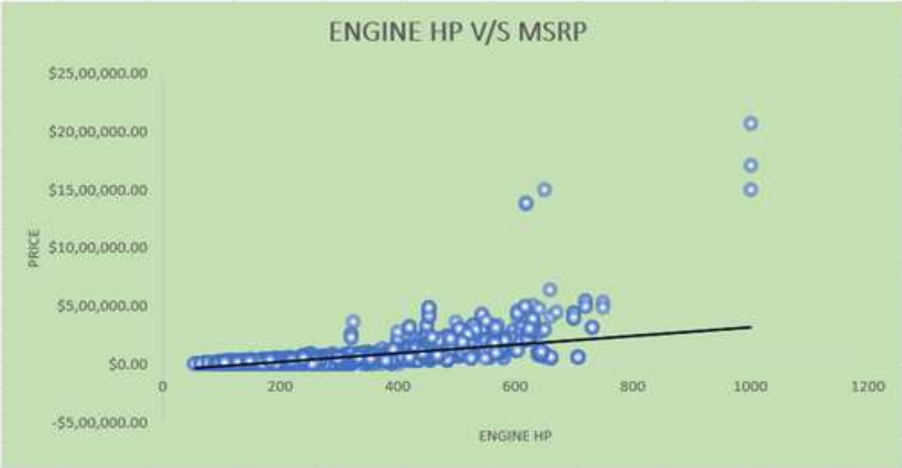
Popularity of a Car Model Vary Across Different Market Categories

Row Labels	Count of Model	Average of Popularity
Flex Fuel,Diesel	16	5657
Hatchback,Flex Fuel	7	5657
Crossover,Flex Fuel,Performance	6	5657
Crossover,Luxury,Performance,Hybrid	2	3916
Crossover,Factory Tuner,Luxury,Performance	5	2607.4
Crossover,Performance	69	2585.956522
Crossover,Hybrid	42	2563.380952
Diesel,Luxury	47	2416.106383
Luxury,Performance,Hybrid	11	2333.181818
Flex Fuel	855	2225.71345
Crossover,Luxury,Diesel	33	2195.848485
Hatchback,Factory Tuner,Performance	21	2173.714286
Factory Tuner,Luxury,High-Performance	215	2133.367442
Hybrid	121	2116.586777
Hatchback,Hybrid	64	2111.15625
Crossover,Flex Fuel	64	2073.75
Crossover,Hatchback,Performance	6	2009
Crossover,Hatchback,Factory Tuner,Performance	6	2009
Factory Tuner,High-Performance	104	1966.442308
Crossover,Factory Tuner,Luxury,High-Performance	26	1823.461538
High-Performance	198	1823.378788
Factory Tuner,Performance	81	1818.049383
Diesel	84	1730.904762
Flex Fuel,Performance	81	1702.358025
Crossover,Hatchback	72	1675.694444
Luxury,High-Performance	334	1668.017964
Hatchback,Luxury,Performance	36	1632.25
Crossover	4430	1629.943792
Crossover,Flex Fuel,Luxury,Performance	6	1624
Performance	503	1443.234592
Factory Tuner,Luxury,Performance	31	1413.419355
Flex Fuel,Luxury,Performance	28	1380.071429
Crossover,Luxury,Performance	112	1349.089286
Hatchback,Luxury	45	1323.133333
Luxury,Performance	659	1293.062215



What is the relationship between a car's engine power and its price?

Engine HP	MSRP
335	$46,135.00
300	$40,650.00
300	$36,350.00
230	$29,450.00
230	$34,500.00
230	$31,200.00
300	$44,100.00
300	$39,300.00
230	$36,900.00
230	$37,200.00
300	$39,600.00
230	$31,500.00
300	$44,400.00
230	$37,200.00
320	$48,250.00
320	$43,550.00
172	$2,000.00
172	$2,000.00
172	$2,000.00
172	$2,000.00
172	$2,000.00
172	$2,000.00
172	$2,000.00
172	$2,000.00
172	$2,000.00
172	$2,000.00
172	$2,000.00
160	$27,495.00
160	$24,995.00



Which car features are most important in determining a car's price?

Engine HP	Engine Cylinders	Number of Door	highway MPG	city mpg	Popularity	MSRP
335	6	2	26	19	3916	$46,135.00
300	6	2	28	19	3916	$40,650.00
300	6	2	28	20	3916	$36,350.00
230	6	2	28	18	3916	$29,450.00
230	6	2	28	18	3916	$34,500.00
230	6	2	28	18	3916	$31,200.00
300	6	2	26	17	3916	$44,100.00
300	6	2	28	20	3916	$39,300.00
230	6	2	28	18	3916	$36,900.00
230	6	2	27	18	3916	$37,200.00
300	6	2	28	20	3916	$39,600.00
230	6	2	28	19	3916	$31,500.00
300	6	2	28	19	3916	$44,400.00
230	6	2	28	19	3916	$37,200.00
320	6	2	25	18	3916	$48,250.00
320	6	2	28	20	3916	$43,550.00
172	6	4	24	17	3105	$2,000.00
172	6	4	20	16	3105	$2,000.00
172	6	4	21	16	3105	$2,000.00
172	6	4	24	17	3105	$2,000.00
172	6	4	20	16	3105	$2,000.00
172	6	4	21	16	3105	$2,000.00
172	6	4	21	16	3105	$2,000.00
172	6	4	22	16	3105	$2,000.00
172	6	4	22	17	3105	$2,000.00
172	6	4	22	16	3105	$2,000.00
172	6	4	21	16	3105	$2,000.00

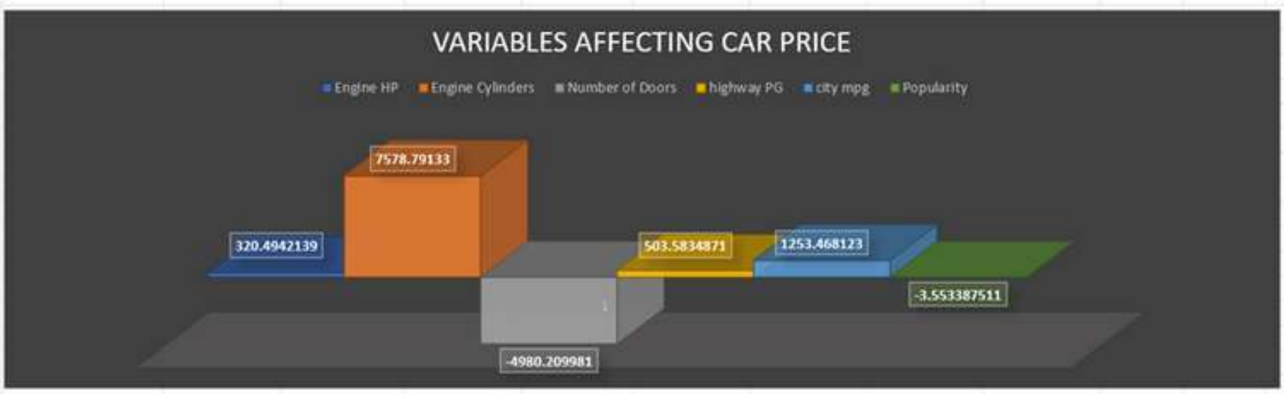
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.683387437
R Square	0.46701839
Adjusted R Square	0.466730032
Standard Error	45078.99614
Observations	11097

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	6	1.9747E+13	3.29117E+12	1619.57869	0
Residual	11090	2.25362E+13	2032115893		
Total	11096	4.22832E+13			

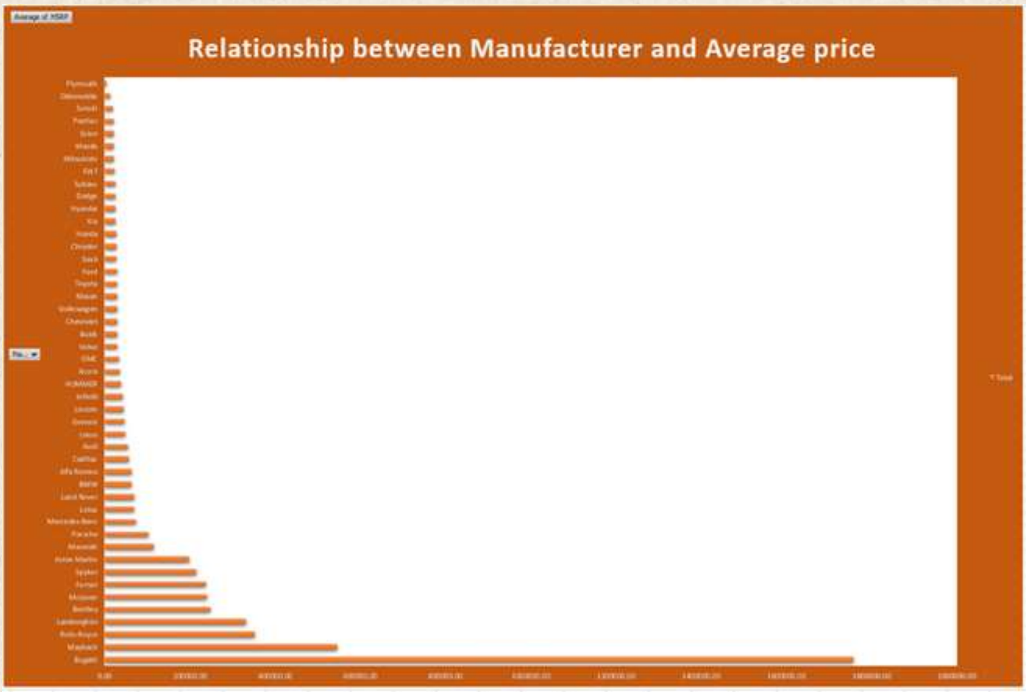
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-97167.8727	3898.078612	-24.9271199	1.714E-133	-104808.8	-89526.9451	-104808.8	-89526.9451
Engine HP	320.4942139	6.3774589	50.25421863	0	307.9932598	332.995168	307.9932598	332.995168
Engine Cylinders	7578.79133	461.2602827	16.43061762	5.8865E-60	6674.639109	8482.94355	6674.639109	8482.94355
Number of Doors	-4980.20998	496.4047724	-10.0325586	1.382E-23	-5953.25165	-4007.16831	-5953.25165	-4007.16831
highway PG	503.5834871	109.2773107	4.608307836	4.1049E-06	289.3805157	717.7864585	289.3805157	717.7864585
city mpg	1253.468123	125.6629389	9.974843293	2.4629E-23	1007.146405	1499.789841	1007.146405	1499.789841
Popularity	-3.55338751	0.297352947	-11.9500666	1.0299E-32	-4.13625219	-2.97052283	-4.13625219	-2.97052283



How does the average price of a car vary across different manufacturers?

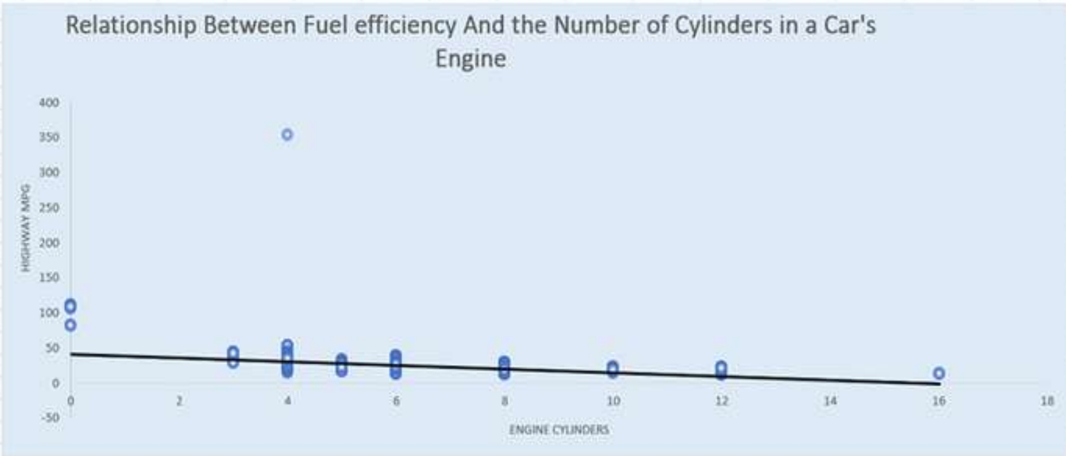
Average Price of a Car Vary Across Different Manufacturers

Manufact	Average of MSRP
Bugatti	1757223.67
Maybach	546221.88
Rolls-Royce	351130.65
Lamborghini	331567.31
Bentley	247169.32
McLaren	239805.00
Ferrari	237383.82
Spyker	214990.00
Aston Martin	198123.46
Maserati	113684.49
Porsche	101622.40
Mercedes-Be	72135.03
Lotus	68377.14
Land Rover	68067.09
BMW	62162.56
Alfa Romeo	61600.00
Cadillac	56368.27
Audi	54574.12
Lexus	47549.07
Genesis	46616.67
Lincoln	43560.01
Infiniti	42640.27



What is the relationship between fuel efficiency and the number of cylinders in a car's engine?

Engine Cylinders	highway MPG
6	26
6	28
6	28
6	28
6	28
6	28
6	26
6	28
6	28
6	27
6	28
6	28
6	28
6	28
6	25
6	28
6	24
6	20
6	21
6	24
6	20
6	21
6	21
6	22

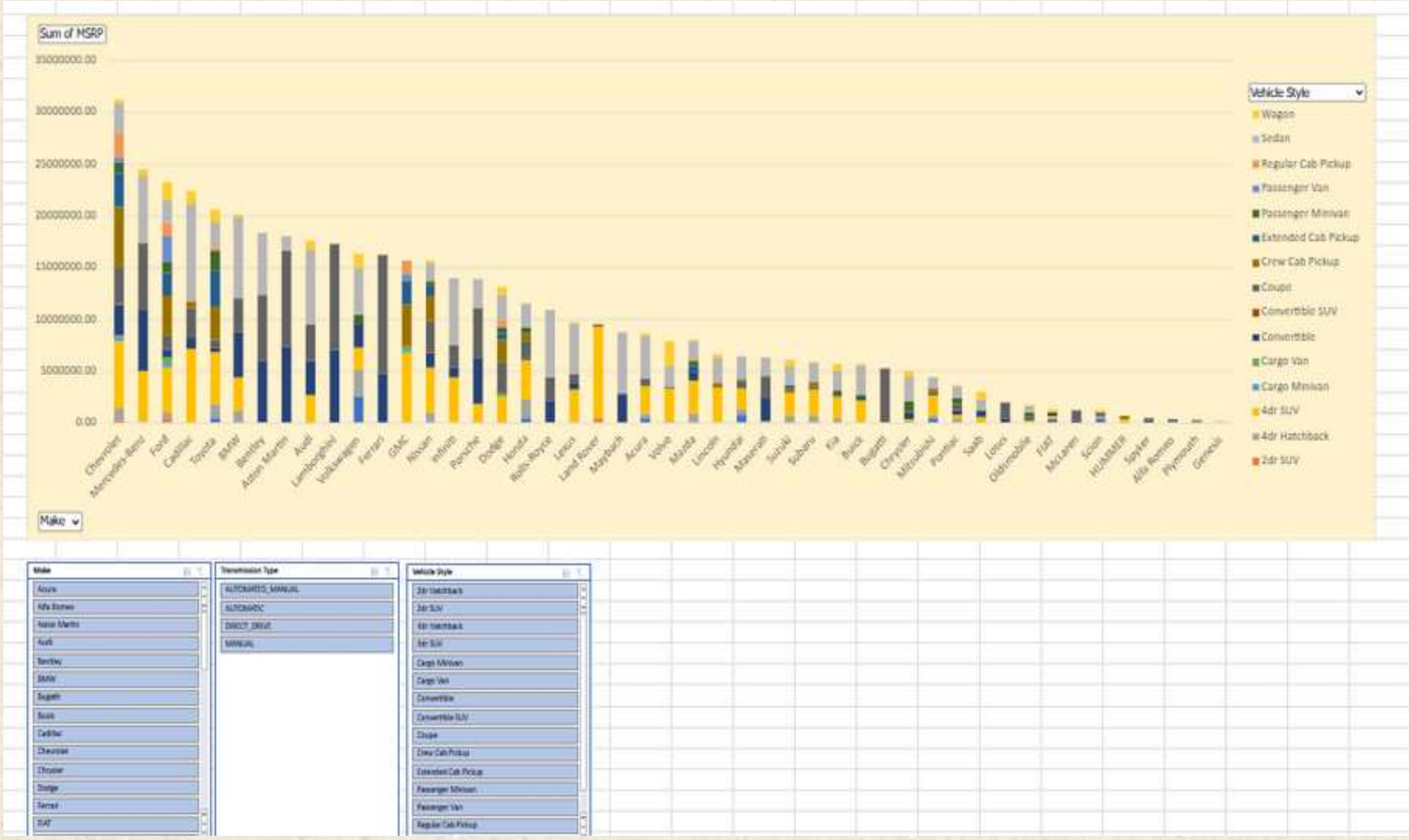


How does the distribution of car prices vary by brand and body style?

DISTRIBUTION OF CAR PRICES BY BRAND AND BODY STYLE

Sum of MSRP	Column Labels							
Row Labels	2dr Hatchback	2dr SUV	4dr Hatchback	4dr SUV	Cargo Minivan	Cargo Van	Convertible	Convertible SUV
Chevrolet	8000.00	193310.00	1209735.00	6509468.00	420150.00	74688.00	2953245.00	106300.00
Mercedes-Benz			122800.00	4924810.00	28950.00		5753964.00	
Ford	24000.00	467873.00	480155.00	4370871.00	394000.00	556351.00	730007.00	
Cadillac				7182555.00			985607.00	
Toyota	473750.00		1397750.00	4957050.00			386668.00	
BMW	80097.00		1103100.00	3160950.00			4403171.00	
Bentley							6012870.00	
Aston Martin							7321655.00	
Audi	4000.00			2674900.00			3291405.00	
Lamborghini							7064450.00	
Volkswagen	2606540.00		2566055.00	2084955.00			2296916.00	
Ferrari							4723811.00	
GMC		128319.00		6633919.00	142750.00	460085.00		
Nissan	14683.00		1023090.00	4149630.00	128620.00		1406552.00	131075.00
Infiniti				4340200.00			980050.00	
Porsche	28827.00			1815200.00			4504586.00	
Dodge	38000.00	12000.00	16000.00	2462875.00	60520.00	338497.00	6000.00	
Honda	413200.00		1846010.00	3800589.00			252135.00	
Rolls-Royce							2141365.00	
Lexus			94700.00	3152974.00			472065.00	
Land Rover		476394.00		8839200.00				145731.00
Maybach							2762750.00	
Acura	480917.00		357440.00	2663505.00				
Volvo	157550.00			3131700.00			121600.00	
Mazda	18000.00	12000.00	853180.00	3175515.00			870505.00	
Lincoln				3422570.00				
Hyundai	789650.00		528880.00	1994390.00				
Maserati				155000.00			2342963.00	
Suzuki	44496.00	12000.00	584387.00	2303493.00				120194.00
Subaru	12000.00		678060.00	2539900.00				
Kia			406960.00	2049645.00				
Buick				2141770.00			179325.00	
Bugatti								
Chrysler	98805.00			250545.00			630105.00	
Mitsubishi	370169.00		334850.00	2009807.00	2000.00		209893.00	
Pontiac	163505.00		162975.00	401550.00			473481.00	
Saab	12000.00		34586.00	341905.00			632628.00	
Lotus							413260.00	
Oldsmobile				238150.00			2000.00	
FIAT	325315.00			369305.00			327965.00	
McLaren							280225.00	
Scion	366325.00		282470.00					
HUMMER				377490.00				

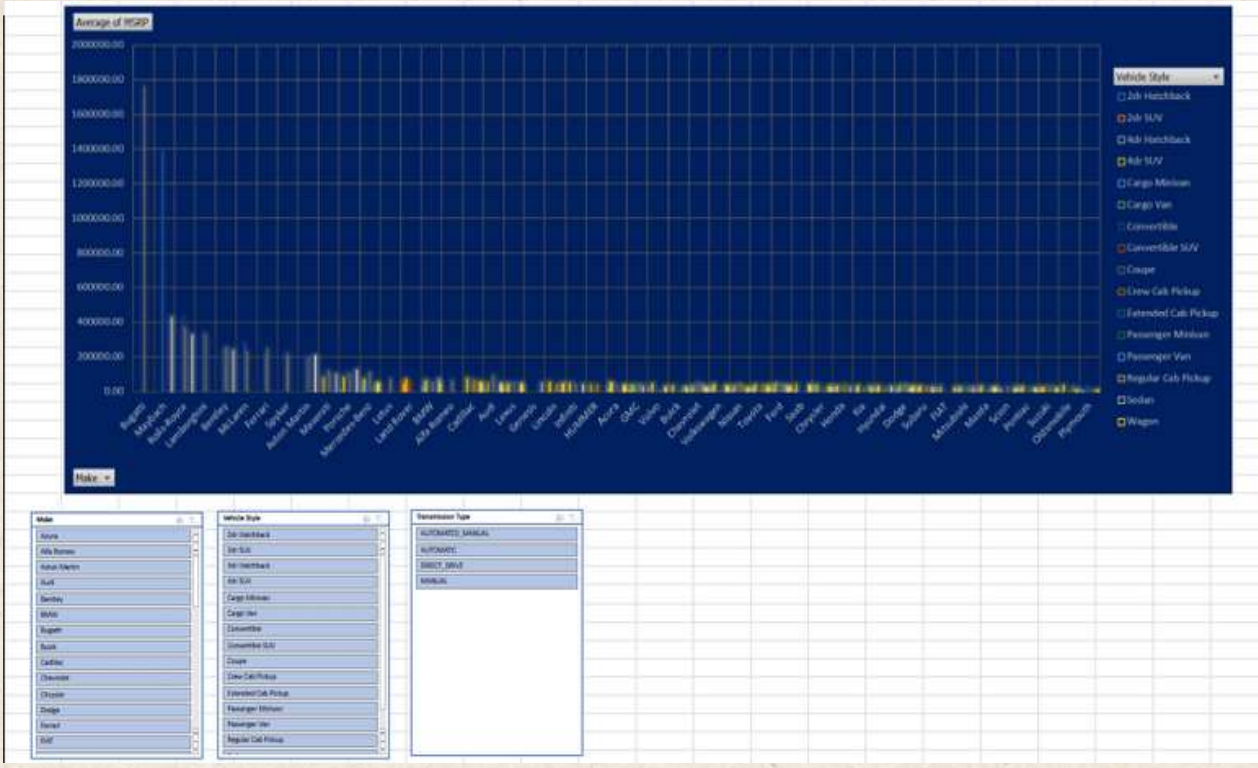
Row Labels	Coupe	Crew Cab Pickup	Extended Cab Pickup	Passenger Minivan	Passenger Van	Regular Cab Pickup	Sedan	Wagon	Grand Total
Chevrolet	3504525.00	5927617.00	3117951.00	1047240.00	599670.00	2260032.00	2942632.00	300675.00	31175238.00
Mercedes-Benz	6473107.00			32500.00			6543743.00	646035.00	24525909.00
Ford	1398144.00	3782518.00	2285584.00	1039010.00	2429898.00	1299240.00	2279348.00	1623565.00	23160564.00
Cadillac	2953574.00	599150.00					9416847.00	1184100.00	22321833.00
Toyota	811995.00	3131895.00	3491424.00	1952518.00		369446.00	2380826.00	1237955.00	20591277.00
BMW	3304051.00						7829700.00	259600.00	20140669.00
Bentley	6356760.00						5920900.00		18290530.00
Aston Martin	9258845.00						1448735.00		18029235.00
Audi	3556290.00						7144348.00	847350.00	17518293.00
Lamborghini	10177050.00								17241500.00
Volkswagen	6000.00			906430.00			4434595.00	1424825.00	16326316.00
Ferrari	11418289.00								16142100.00
GMC		4062482.00	2175866.00	150630.00	599670.00	1284328.00			15638049.00
Nissan	2937632.00	2422300.00	1026379.00	413320.00		19914.00	1763130.00	175000.00	15611325.00
Infiniti	2175750.00						6490009.00		13986009.00
Porsche	4758533.00						2713500.00		13820646.00
Dodge	2973842.00	2072780.00	684682.00	557425.00	70708.00	653408.00	2409585.00	793055.00	13149377.00
Honda	1588705.00	750215.00		553185.00			2264390.00		11468429.00
Rolls-Royce	2204675.00						6539010.00		10885050.00
Lexus	1016472.00						4837596.00	31105.00	9604912.00
Land Rover									9461325.00
Maybach							5976800.00		8739550.00
Acura	793748.00						4134552.00	201360.00	8631522.00
Volvo	6000.00						2072945.00	2416971.00	7906766.00
Mazda	12000.00		580033.00	443130.00		265486.00	1618571.00	33350.00	7881770.00
Lincoln	17342.00	453260.00					2458245.00	269705.00	6621122.00
Hyundai	685920.00			133075.00			2323987.00		6455902.00
Maserati	1972284.00						1782400.00		6252647.00
Suzuki		304131.00	259659.00				1797070.00	683707.00	6109137.00
Subaru	354476.00	365975.00					1833110.00	10000.00	5793521.00
Kia	142630.00			494650.00			1976360.00	601155.00	5671400.00
Buick	18534.00			330065.00			2838590.00	8212.00	5516496.00
Bugatti	5271671.00								5271671.00
Chrysler	114510.00			922295.00			2479859.00	501075.00	4997194.00
Mitsubishi		240210.00	134360.00	2000.00		8000.00	1058563.00		4369852.00
Pontiac	663715.00			541192.00			1156535.00	20855.00	3583808.00
Saab							1066500.00	751280.00	3038899.00
Lotus	1501300.00								1914560.00
Oldsmobile	276015.00			492055.00			667161.00	20000.00	1695381.00
FIAT								287570.00	1310155.00
McLaren	918800.00								1199025.00
Scion	330210.00						32500.00	184445.00	1195950.00
HUMMER		242405.00							619895.00



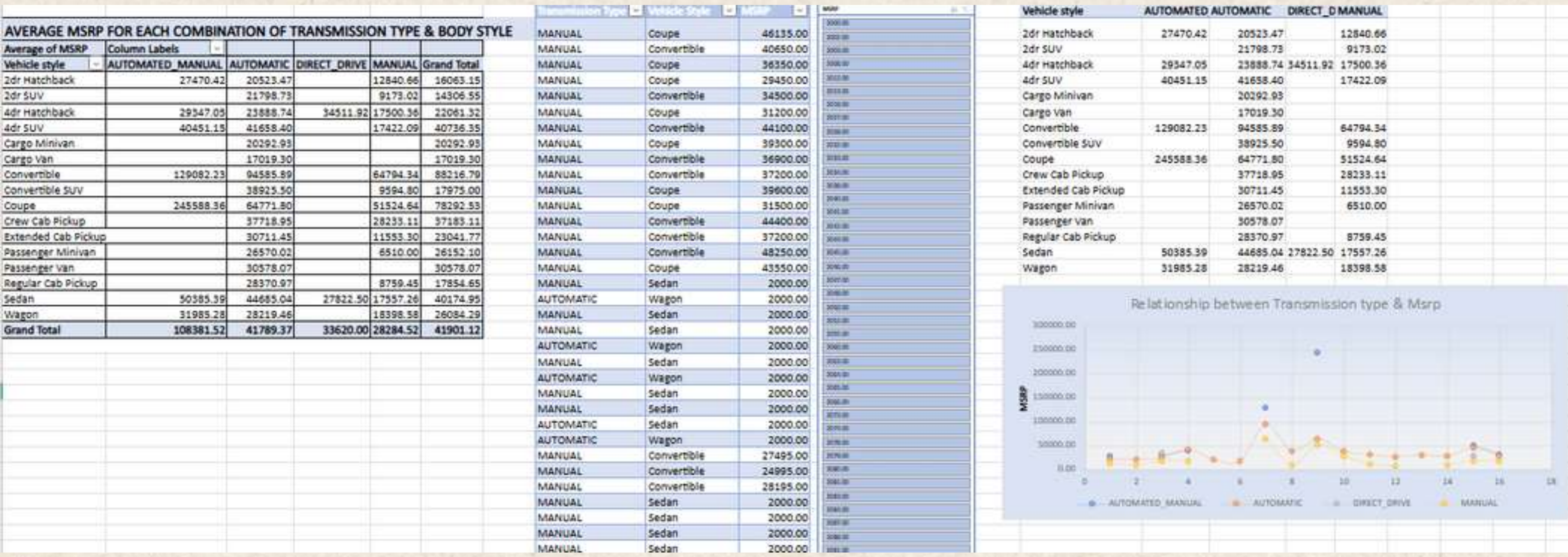
Which car brands have the highest and lowest average MSRPs, and how does this vary by body style?

AVERAGE MSRPs ACROSS DIFFERENT CAR BRANDS AND BODY STYLES

Average of MSRP	Body style																
Row Labels	2dr Hatchback	2dr SUV	4dr Hatchback	4dr SUV	Cargo Minivan	Cargo Van	Convertible	Convertible SUV	Coupe	Crew Cab Pickup	Extended Cab Pickup	Passenger Minivan	Passenger Van	Regular Cab Pickup	Sedan	Wagon	Grand Total
Bugatti									1757223.67								1757223.67
Maybach							1381375.00								426914.29		546221.88
Rolls-Royce							428273.00		367445.83						326950.50		351130.65
Lamborghini							336402.38		328291.94								331567.31
Bentley							250536.25		254270.40						236836.00		247169.32
McLaren							280225.00		229700.00								239805.00
Ferrari							214718.68		248223.67								237383.82
Spyker							219990.00		209990.00								214990.00
Aston Martin							203379.31		192892.60						206962.14		198123.46
Maserati				77500.00			130164.61		116016.71						99022.22		113684.49
Porsche	5765.40			82509.09			115502.21		99136.10						123340.91		101622.40
Mercedes-Benz			40933.33	68400.14	28950.00		104617.53		109715.68			32500.00			48833.90	43069.00	72135.03
Lotus							51657.50		75065.00								68377.14
Land Rover		39699.50		71283.87				48577.00									68067.09
BMW	26699.00		55155.00	58536.11			63814.07		52445.25						71832.11	43266.67	62162.56
Alfa Romeo							64900.00		59400.00								61600.00
Cadillac				72551.06			70400.50		45439.60	66572.22					51178.52	47364.00	56368.27
Audi	2000.00			48634.55			70029.89		93586.58						46391.87	33894.00	54574.12
Lexus			31566.67	45042.49			52451.67		50823.60						48864.61	31105.00	47549.07
Genesis															46616.67		46616.67
Lincoln				50331.91					2167.75	41205.45					41665.17	44950.83	43560.01
Infiniti				45686.32			46669.05		40291.67						41076.01		42640.27
HUMMER				37749.00						34629.29							36464.41
Acura	17175.61		51062.86	42959.76					39687.40						33614.24	33560.00	35087.49
GMC		7128.83		37479.77	23791.67	21908.81				39062.33	27895.72	25105.00	28555.71	25182.90			32444.09
Volvo	26258.33			45386.96			40533.33		2000.00						22289.73	26271.42	29724.68
Buick				33996.35			25617.86		2059.33			30005.91			29568.65	2053.00	29034.19
Chevrolet	2000.00	13807.86	18329.32	33553.96	20007.14	8298.67	62835.00	17716.67	38939.17	39255.74	24170.16	24934.29	28555.71	19824.84	19882.65	15825.00	29000.22
Volkswagen	24134.63		28198.41	41699.10			27673.69		2000.00			29239.68			30795.80	26385.65	28947.37
Nissan	2097.57		22241.09	34294.46	21436.67		39070.89	43691.67	35393.16	32733.78	20527.58	22962.22		2212.67	22604.23	17500.00	28856.42
Toyota	18950.00		22186.51	40631.36			25777.87		15615.29	36845.82	26251.31	30038.74		17592.67	24800.27	31742.44	28758.77
Ford	2000.00	16133.53	18467.50	42027.61	19700.00	20605.59	34762.24		34101.07	41566.13	23808.17	22587.17	32836.46	17797.81	23258.65	30066.02	28522.86
Saab	2000.00		2034.47	41685.00			28755.82								36775.86	34149.09	27879.81
Chrysler	32935.00			35792.14			24234.81		19085.00			29751.45			26103.78	26372.37	26722.96
Honda	17216.67		26371.57	28575.86			36019.29		21763.08	34100.68		36879.00			26027.47		26608.88
Kia			19379.05	31533.00					20375.71			32976.67			23811.57	18216.82	25318.75
Hyundai	18363.95		17629.33	30218.03					22126.45			26615.00			27666.51		24926.26
Dodge	2000.00	2000.00	2000.00	31175.63	20173.33	12536.93	2000.00		45058.21	31405.76	16301.95	25337.50	14141.60	14850.18	22519.49	24782.97	24857.05
Subaru	2000.00		21189.38	28538.20					16112.55	24398.33					26187.29	2000.00	24240.67
FIAT	19136.18			24620.33			23426.07									22120.77	22206.02
Mitsubishi	12764.45		12878.85	26101.39	2000.00		29984.71			26690.00	19194.29	2000.00		2000.00	24058.25		21316.35
Mazda	2000.00	2000.00	20809.27	27141.15			28080.81		2000.00		11600.66	23322.63		9154.69	19738.67	16675.00	20106.56
Scion	20351.39		15692.78						27517.50						16250.00	18444.50	19932.50



How do the different feature such as transmission type affect the MSRP, and how does this vary by body style?

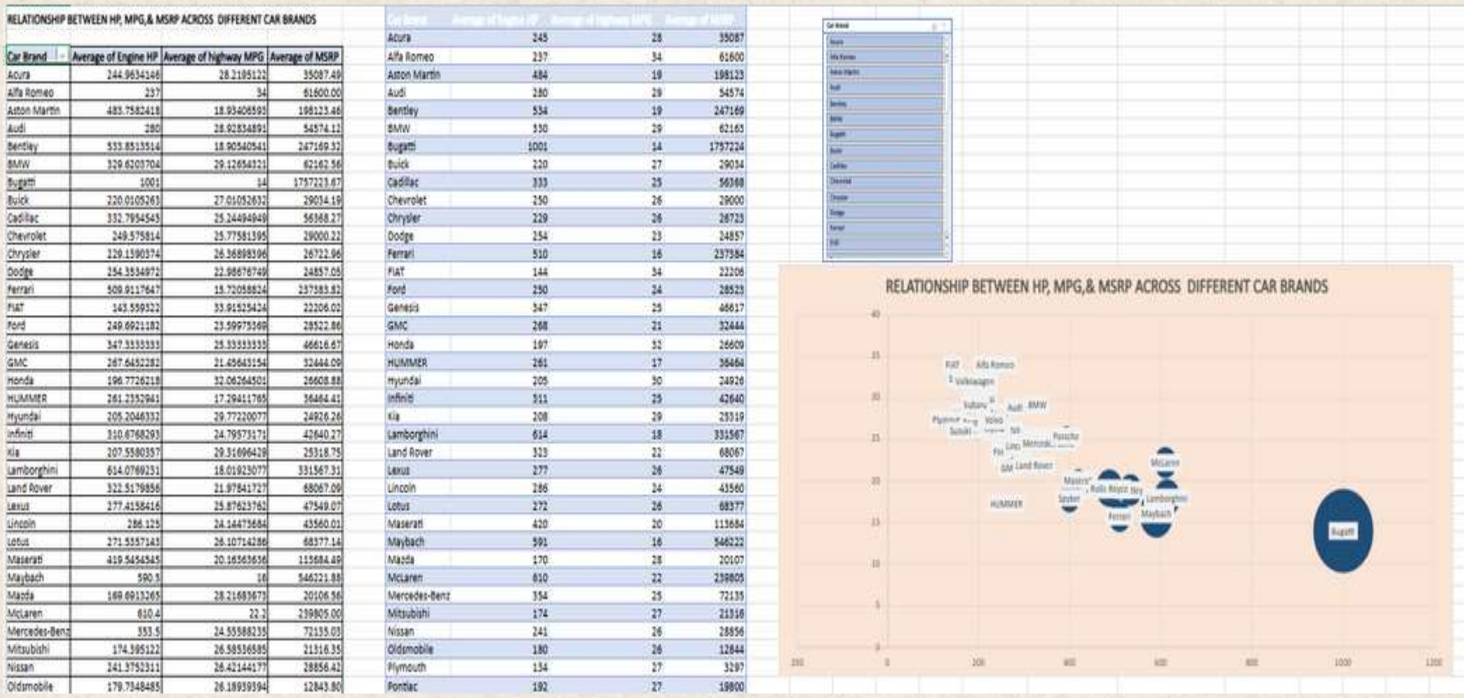


AVERAGE MSRP FOR EACH COMBINATION OF TRANSMISSION TYPE & BODY STYLE

Average of MSRP	Column Labels				
Vehicle style	AUTOMATED_MANUAL	AUTOMATIC	DIRECT_DRIVE	MANUAL	Grand Total
2dr Hatchback	27470.42	20523.47		12840.66	16063.15
2dr SUV		21798.73		9173.02	14306.55
4dr Hatchback	29347.05	23888.74	34511.92	17500.36	22061.32
4dr SUV	40451.15	41658.40		17422.09	40736.35
Cargo Minivan		20292.93			20292.93
Cargo Van		17019.30			17019.30
Convertible	129082.23	94585.89		64794.34	88216.79
Convertible SUV		38925.50		9594.80	17975.00
Coupe	245588.36	64771.80		51524.64	78292.53
Crew Cab Pickup		37718.95		28233.11	37183.11
Extended Cab Pickup		30711.45		11553.30	23041.77
Passenger Minivan		26570.02		6510.00	26152.10
Passenger Van		30578.07			30578.07
Regular Cab Pickup		28370.97		8759.45	17854.65
Sedan	50385.39	44685.04	27822.50	17557.26	40174.95
Wagon	31985.28	28219.46		18398.58	26084.29
Grand Total	108381.52	41789.37	33620.00	28284.52	41901.12

How does the fuel efficiency of cars vary across different body styles and model years?

How does the car's horsepower, MPG, and price vary across different Brands?



Car Brand	Average of Engine HP	Average of highway MPG	Average of MSRP
Acura	244.9634146	28.2195122	35087.49
Alfa Romeo	237	34	61600.00
Aston Martin	483.7582418	18.93406593	198123.46
Audi	280	28.92834891	54574.12
Bentley	533.8513514	18.90540541	247169.32
BMW	329.6203704	29.12654321	62162.56
Bugatti	1001	14	1757223.67
Buick	220.0105263	27.01052632	29034.19
Cadillac	332.7954545	25.24494949	56368.27
Chevrolet	249.575814	25.77581395	29000.22
Chrysler	229.1590374	26.36898396	26722.96
Dodge	254.3534972	22.98676749	24857.05
Ferrari	509.9117647	15.72058824	237383.82
FIAT	143.559322	33.91525424	22206.02
Ford	249.6921182	23.59975369	28522.86
Genesis	347.3333333	25.33333333	46616.67
GMC	267.6452282	21.45643154	32444.09
Honda	196.7726218	32.06264501	26608.88
HUMMER	261.2352941	17.29411765	36464.41
Hyundai	205.2046332	29.77220077	24926.26
Infiniti	310.6768293	24.79573171	42640.27
Kia	207.5580357	29.31696429	25318.75
Lamborghini	614.0769231	18.01923077	331567.31
Land Rover	322.5179856	21.97841727	68067.09
Lexus	277.4158416	25.87623762	47549.07
Lincoln	286.125	24.14473684	43560.01
Lotus	271.5357143	26.10714286	68377.14
Maserati	419.5454545	20.16363636	113684.49
Maybach	590.5	16	546221.88
Mazda	169.6913265	28.21683673	20106.56
McLaren	610.4	22.2	239805.00
Mercedes-Benz	353.5	24.55588235	72135.03
Mitsubishi	174.395122	26.58536585	21316.35
Nissan	241.3752311	26.42144177	28856.42
Oldsmobile	179.7348485	26.18939394	12843.80

Car Brand	Average of Engine HP	Average of highway MPG	Average of MSRP
Acura	245	28	35087
Alfa Romeo	237	34	61600
Aston Martin	484	19	198123
Audi	280	29	54574
Bentley	534	19	247169
BMW	330	29	62163
Bugatti	1001	14	1757224
Buick	220	27	29034
Cadillac	333	25	56368
Chevrolet	250	26	29000
Chrysler	229	26	26723
Dodge	254	23	24857
Ferrari	510	16	237384
FIAT	144	34	22206
Ford	250	24	28523
Genesis	347	25	46617
GMC	268	21	32444
Honda	197	32	26609
HUMMER	261	17	36464
Hyundai	205	30	24926
Infiniti	311	25	42640
Kia	208	29	25319
Lamborghini	614	18	331567
Land Rover	323	22	68067
Lexus	277	26	47549
Lincoln	286	24	43560
Lotus	272	26	68377
Maserati	420	20	113684
Maybach	591	16	546222
Mazda	170	28	20107
McLaren	610	22	239805
Mercedes-Benz	354	25	72135
Mitsubishi	174	27	21316
Nissan	241	26	28856
Oldsmobile	180	26	12844
Plymouth	134	27	3297
Pontiac	192	27	19800

Conclusion:

Predicting the price of a car based on its features and market category: By using the various features and market category variables in the dataset, a data analyst could develop a model to predict the price of a car. This could help manufacturers and consumers understand how different features affect the price of a car and make informed decisions about pricing and purchasing.

ABC Call Volume Trend

Project Description:

In this project, you'll be diving into the world of Customer Experience (CX) analytics, specifically focusing on the inbound calling team of a company. You'll be provided with a dataset that spans 23 days and includes various details such as the agent's name and ID, the queue time (how long a customer had to wait before connecting with an agent), the time of the call, the duration of the call, and the call status (whether it was abandoned, answered, or transferred).

Problem:

The focus of this project, involves handling incoming calls from existing or prospective customers. The goal is to attract, engage, and delight customers, turning them into loyal advocates for the business.

Design:

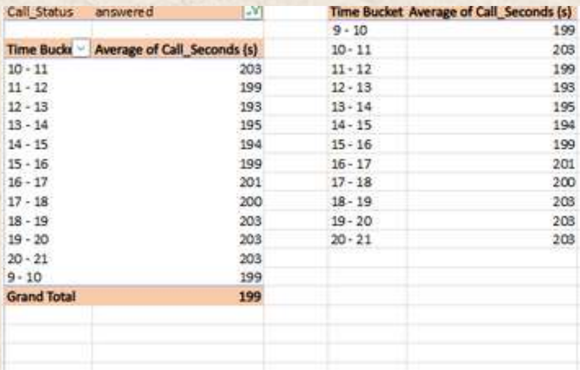
Steps taken to clean the data

- First importing the datasets provided and merging them.
- Then removing the duplicates and the blank cells.
- Improving the headers of each column with proper values.
- Finding and replacing the data with correct values.

Tools used for visualization: Microsoft Excel

Findngs:

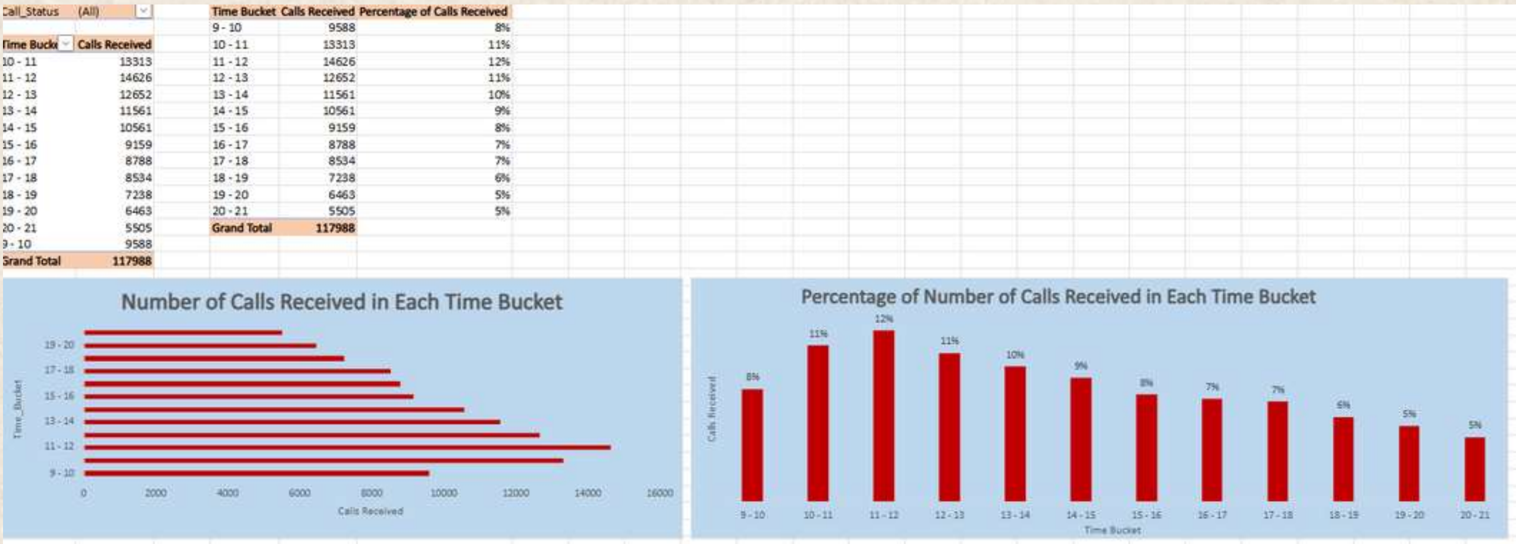
Average Call Duration: Determine the average duration of all incoming calls received by agents. This should be calculated for each time bucket.



Call_Status	answered
Time Bucket	Average of Call_Seconds (s)
10 - 11	203
11 - 12	199
12 - 13	193
13 - 14	195
14 - 15	194
15 - 16	199
16 - 17	201
17 - 18	200
18 - 19	203
19 - 20	203
20 - 21	203
9 - 10	199
Grand Total	199

Time Bucket	Average of Call_Seconds (s)
9 - 10	199
10 - 11	203
11 - 12	199
12 - 13	193
13 - 14	195
14 - 15	194
15 - 16	199
16 - 17	201
17 - 18	200
18 - 19	203
19 - 20	203
20 - 21	203

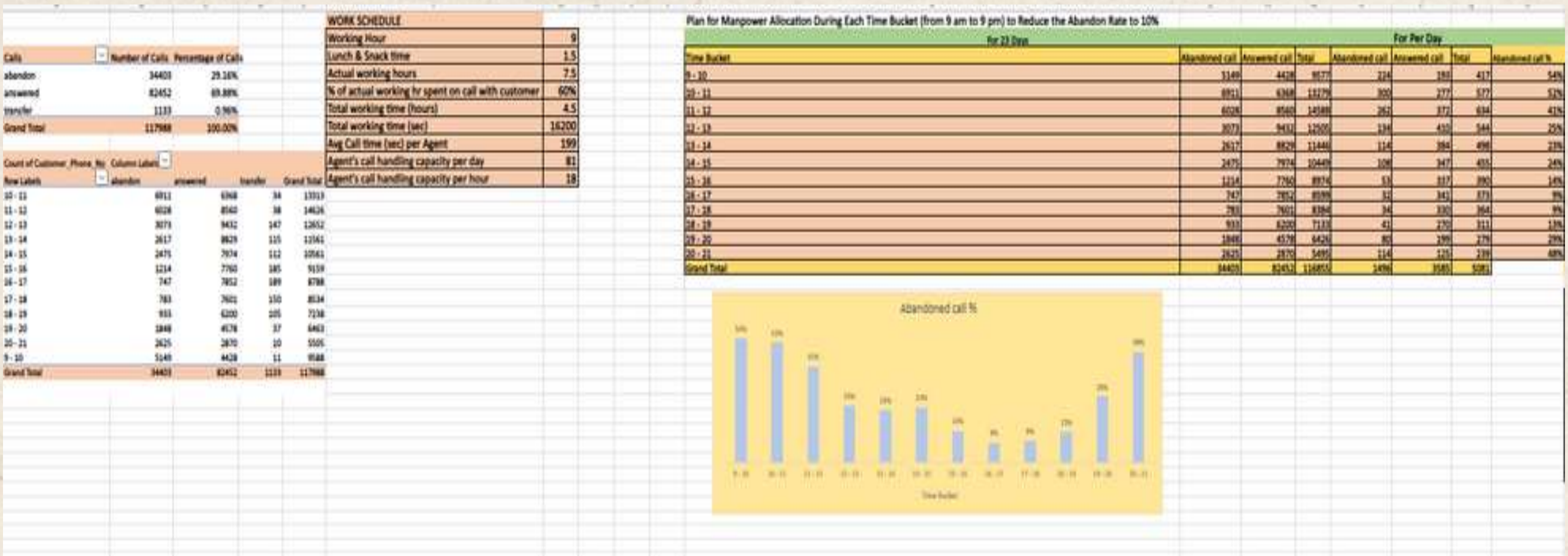
Call Volume Analysis: Visualize the total number of calls received. This should be represented as a graph or chart showing the number of calls against time. Time should be represented in buckets (e.g., 1-2, 2-3, etc.).



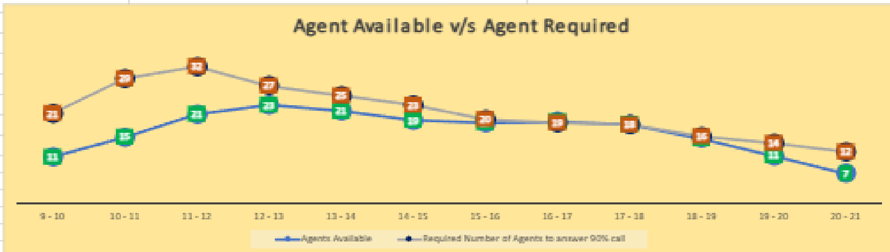
Call_Status	(All)
Time Bucket	Calls Received
10 - 11	13313
11 - 12	14626
12 - 13	12652
13 - 14	11561
14 - 15	10561
15 - 16	9159
16 - 17	8788
17 - 18	8534
18 - 19	7238
19 - 20	6463
20 - 21	5505
9 - 10	9588
Grand Total	117988

Time Bucket	Calls Received	Percentage of Calls Received
9 - 10	9588	8%
10 - 11	13313	11%
11 - 12	14626	12%
12 - 13	12652	11%
13 - 14	11561	10%
14 - 15	10561	9%
15 - 16	9159	8%
16 - 17	8788	7%
17 - 18	8534	7%
18 - 19	7238	6%
19 - 20	6463	5%
20 - 21	5505	5%
Grand Total	117988	

Manpower Planning: The current rate of abandoned calls is approximately 30%. Propose a plan for manpower allocation during each time bucket (from 9 am to 9 pm) to reduce the abandon rate to 10%. In other words, you need to calculate the minimum number of agents required in each time bucket to ensure that at least 90 out of 100 calls are answered.



Required Daily Call Volume to Achieve 90% Performance
375
519
571
490
448
410
351
336
328
280
251
215
4573



Time Bucket	Answered call	Agents Available	Required Daily Call Volume to Achieve 90% Performance	Required Number of Agents to answer 90% call
9 - 10	193	11	375	21
10 - 11	277	15	519	29
11 - 12	372	21	571	32
12 - 13	410	23	490	27
13 - 14	384	21	448	25
14 - 15	347	19	410	23
15 - 16	337	19	351	20
16 - 17	341	19	336	19
17 - 18	330	18	328	18
18 - 19	270	15	280	16
19 - 20	199	11	251	14
20 - 21	125	7	215	12
	3585	199	4573	254

Night Shift Manpower Planning: Customers also call ABC Insurance Company at night but don't get an answer because there are no agents available. This creates a poor customer experience. Assume that for every 100 calls that customers make between 9 am and 9 pm, they also make 30 calls at night between 9 pm and 9 am. The distribution of these 30 calls is as follows:

Time Bucket	Required Daily Call Volume to Achieve 90% Performance
9 - 10	375
10 - 11	519
11 - 12	571
12 - 13	490
13 - 14	448
14 - 15	410
15 - 16	351
16 - 17	336
17 - 18	328
18 - 19	280
19 - 20	251
20 - 21	215
Total No of Calls Per Day	4573
Total No of Calls Per Night	1372

Time Bucket	Distribution of 30 calls	Percentage of Distribution of 30 Calls	No of Calls	No of Agent Required
9 pm-10 pm	3	10%	137	8
10 pm-11 pm	3	10%	137	8
11 pm-12 am	2	7%	91	5
12 am-01 am	2	7%	91	5
01 am-02 am	1	3%	46	3
02 am-03 am	1	3%	46	3
03 am-04 am	1	3%	46	3
04 am-05 am	1	3%	46	3
05 am-06 am	3	10%	137	8
06 am-07 am	4	13%	183	10
07 am-08 am	4	13%	183	10
08 am-09 am	5	17%	229	13
Total	30	100%	1372	76

Time Bucket	Required Number of Agents to answer 90% call
9 - 10	21
10 - 11	29
11 - 12	32
12 - 13	27
13 - 14	25
14 - 15	23
15 - 16	20
16 - 17	19
17 - 18	18
18 - 19	16
19 - 20	14
20 - 21	12
21 - 22	8
22 - 23	8
23 - 24	5
24 - 01	5
01 - 02	3
02 - 03	3
03 - 04	3
04 - 05	3
05 - 06	8
06 - 07	10
07 - 08	10
08 - 09	13
Total	330

Plan for Each Time Bucket Throughout the Day & Night

Conclusion:

In this project, I'll be using your analytical skills to understand the trends in the call volume of the CX team and derive valuable insights from it.

Appendix:

Instagram User Analytics:

[Instagram User Analytics]

Hiring Process Analytics:

[Hiring Process Analytics]

IMDB Movie Analysis:

[IMDB Movie Analysis]

Bank Loan Case Study:

[Bank Loan Case Study]

Impact of Car Features:

[Impact Of Car Features]

ABC Call Volume Trend:

[ABC Call Volume Trend]

Thank You